

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
MTD selection (%)						
U2 - shared alpha	70.9	9.9	2.8	0.2	0	16.2
U2 - dose-specific alpha	70.9	9.8	2.7	0.3	0	16.3
U2 - cycle-1 oracle	65.7	14.2	7.3	2.6	0.7	9.5
U3 - shared alpha	70.8	10.2	2.7	0.3	0	16
U3 - dose-specific alpha	70.8	9.9	2.6	0.4	0	16.3
U3 - cycle-1 oracle	65.3	14.5	7.4	2.6	0.7	9.5
OBD selection (%)						
U2 - shared alpha	13.2	4	0.5	0.1	0	82.2
U2 - dose-specific alpha	13.4	4.2	0.6	0	0	81.8
U2 - cycle-1 oracle	16	7.1	1.7	0.4	0.2	74.6
U3 - shared alpha	14.4	3.4	0.4	0	0	81.8
U3 - dose-specific alpha	13.9	3.3	0.7	0	0	82.1
U3 - cycle-1 oracle	16.4	7.5	1.5	0.4	0.2	74
Mean administrations						
U2 - shared alpha	11.5	2.9	1.3	0.3	0.1	-
U2 - dose-specific alpha	11.5	3	1.3	0.3	0.1	-
U2 - cycle-1 oracle	11.1	3.6	2.1	0.9	0.3	-
U3 - shared alpha	11.5	2.9	1.3	0.4	0.1	-
U3 - dose-specific alpha	11.5	2.9	1.3	0.3	0.1	-
U3 - cycle-1 oracle	11.1	3.5	2.1	0.9	0.3	-
Mean IPDE administrations						
U2 - shared alpha	3.4	1.6	0.5	0.2	0	-
U2 - dose-specific alpha	3.4	1.6	0.5	0.2	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.4	1.6	0.5	0.2	0	-
U3 - dose-specific alpha	3.4	1.6	0.5	0.2	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						573.9
U2 - dose-specific alpha						574.8
U2 - cycle-1 oracle						986.7
U3 - shared alpha						573.1
U3 - dose-specific alpha						573.7
U3 - cycle-1 oracle						985.8
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
MTD selection (%)						
U2 - shared alpha	28.1	24	30.8	12.5	3.9	0.7
U2 - dose-specific alpha	28.2	22.9	31.6	12.3	4.3	0.7
U2 - cycle-1 oracle	22.7	18.7	30.1	19.3	8.7	0.5
U3 - shared alpha	28.2	23.8	31.2	12	4.2	0.6
U3 - dose-specific alpha	27.9	24.7	29.9	12.6	4.3	0.6
U3 - cycle-1 oracle	22.9	18.8	29.8	18.9	9.1	0.5
OBD selection (%)						
U2 - shared alpha	37.5	20.3	30.3	6.7	1.1	4.1
U2 - dose-specific alpha	37.8	19.8	30.7	6.4	1.2	4.1
U2 - cycle-1 oracle	32.9	16.2	31.5	11.1	3	5.3
U3 - shared alpha	40.5	20.5	28.2	6	0.7	4.1
U3 - dose-specific alpha	40.4	21.3	27.7	6.1	0.5	4
U3 - cycle-1 oracle	35.6	16.5	30.7	9.6	2.2	5.4
Mean administrations						
U2 - shared alpha	12.5	5.6	6.9	3.1	1.1	-
U2 - dose-specific alpha	12.5	5.7	6.8	3.2	1.2	-
U2 - cycle-1 oracle	10.9	5.4	7.1	3.9	1.9	-
U3 - shared alpha	12.5	5.6	6.9	3.1	1.2	-
U3 - dose-specific alpha	12.5	5.7	6.8	3.2	1.2	-
U3 - cycle-1 oracle	10.9	5.2	7.3	3.8	1.9	-
Mean IPDE administrations						
U2 - shared alpha	3.7	2.6	1.9	0.9	0.3	-
U2 - dose-specific alpha	3.7	2.6	1.9	0.9	0.3	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.7	2.6	1.9	0.9	0.3	-
U3 - dose-specific alpha	3.7	2.6	1.9	0.9	0.3	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1094.3
U2 - dose-specific alpha						1094.8
U2 - cycle-1 oracle						1617.8
U3 - shared alpha						1095.5
U3 - dose-specific alpha						1096
U3 - cycle-1 oracle						1617.5

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
MTD selection (%)						
U2 - shared alpha	29	24	29.3	13.4	3.5	0.8
U2 - dose-specific alpha	28.9	24.1	29.4	13.6	3.2	0.8
U2 - cycle-1 oracle	23.7	17.7	30.5	18.4	9.2	0.5
U3 - shared alpha	28.8	23.6	29.9	13.6	3.3	0.8
U3 - dose-specific alpha	29.2	24.2	29.5	12.9	3.4	0.8
U3 - cycle-1 oracle	23.4	17.9	30.7	17.8	9.7	0.5
OBD selection (%)						
U2 - shared alpha	15.2	40.4	19.1	6	0.6	18.7
U2 - dose-specific alpha	15.4	41.3	18.1	6.2	0.6	18.4
U2 - cycle-1 oracle	10.9	40.2	20.4	7.3	2.3	18.9
U3 - shared alpha	16.4	43.2	16.9	4.4	0.5	18.6
U3 - dose-specific alpha	16.9	44.4	14.7	4.9	0.5	18.6
U3 - cycle-1 oracle	11.6	43.2	18.5	5.9	2	18.8
Mean administrations						
U2 - shared alpha	9.4	7.8	5.7	3	1.1	-
U2 - dose-specific alpha	9.4	7.9	5.7	2.9	1.2	-
U2 - cycle-1 oracle	7.8	7.7	6.1	3.6	1.8	-
U3 - shared alpha	9.5	7.9	5.7	2.9	1.1	-
U3 - dose-specific alpha	9.5	7.8	5.8	2.9	1.1	-
U3 - cycle-1 oracle	7.8	7.8	6.1	3.6	1.7	-
Mean IPDE administrations						
U2 - shared alpha	2.7	3.4	1.6	0.9	0.3	-
U2 - dose-specific alpha	2.7	3.4	1.6	0.9	0.3	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.7	3.4	1.6	0.9	0.3	-
U3 - dose-specific alpha	2.7	3.3	1.6	0.9	0.3	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						997.5
U2 - dose-specific alpha						997.4
U2 - cycle-1 oracle						1501.4
U3 - shared alpha						997
U3 - dose-specific alpha						997.1
U3 - cycle-1 oracle						1500.1
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
MTD selection (%)						
U2 - shared alpha	3.3	3.8	29.5	44.3	19.1	0
U2 - dose-specific alpha	3.3	3.8	29.3	43.9	19.7	0
U2 - cycle-1 oracle	2.1	2.8	21.9	38.7	34.5	0
U3 - shared alpha	3.4	3.9	30	43.4	19.3	0
U3 - dose-specific alpha	3.4	4	28.7	44.4	19.5	0
U3 - cycle-1 oracle	2.1	2.9	21	40.7	33.3	0
OBD selection (%)						
U2 - shared alpha	8.4	10.6	24.7	41.7	10.9	3.7
U2 - dose-specific alpha	7.8	10.5	24.7	42	10.9	4.1
U2 - cycle-1 oracle	6.5	7.3	22.6	42.5	18.4	2.7
U3 - shared alpha	12.7	11.5	22.2	40	9.8	3.8
U3 - dose-specific alpha	12.1	11.5	22.3	40.7	9.5	3.9
U3 - cycle-1 oracle	11.5	8.2	19.7	43.6	14.4	2.6
Mean administrations						
U2 - shared alpha	4.5	4.5	7.2	8	5	-
U2 - dose-specific alpha	4.5	4.5	7.2	8.2	4.9	-
U2 - cycle-1 oracle	4.3	4.3	6.7	8.1	6	-
U3 - shared alpha	4.4	4.5	7.3	8.1	4.9	-
U3 - dose-specific alpha	4.5	4.5	7.3	8.1	4.9	-
U3 - cycle-1 oracle	4.3	4.2	6.7	8.2	5.9	-
Mean IPDE administrations						
U2 - shared alpha	0.7	2.9	2.3	2.6	1.2	-
U2 - dose-specific alpha	0.7	2.8	2.3	2.7	1.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.7	2.9	2.3	2.7	1.2	-
U3 - dose-specific alpha	0.7	2.9	2.3	2.7	1.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1077.2
U2 - dose-specific alpha						1076.2
U2 - cycle-1 oracle						1630.7
U3 - shared alpha						1078
U3 - dose-specific alpha						1077.1
U3 - cycle-1 oracle						1630.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
MTD selection (%)						
U2 - shared alpha	7.3	5.2	16.8	30.2	40.4	0.1
U2 - dose-specific alpha	7.2	5.6	15.9	30.5	40.7	0.1
U2 - cycle-1 oracle	5.6	3.3	13.5	23.6	53.9	0.1
U3 - shared alpha	7.2	5.4	16.1	30.3	40.9	0.1
U3 - dose-specific alpha	7.2	5.5	16.3	30.1	40.8	0.1
U3 - cycle-1 oracle	5.6	3.2	13.4	23.8	53.9	0.1
OBD selection (%)						
U2 - shared alpha	3.5	35.7	27.5	19.1	7.1	7.1
U2 - dose-specific alpha	4	35.3	26.7	20	6.9	7.1
U2 - cycle-1 oracle	3.5	37.4	25.3	18.3	10	5.5
U3 - shared alpha	6.2	40.2	25.6	16.6	4.3	7.1
U3 - dose-specific alpha	6.7	40.8	24.8	16.3	4.3	7.1
U3 - cycle-1 oracle	6.6	43.2	23.5	14.6	6.6	5.5
Mean administrations						
U2 - shared alpha	4.6	6.8	6.7	5.9	4.5	-
U2 - dose-specific alpha	4.6	6.7	6.6	6.1	4.6	-
U2 - cycle-1 oracle	4.5	6.6	6.6	5.9	5.3	-
U3 - shared alpha	4.6	6.7	6.7	6	4.5	-
U3 - dose-specific alpha	4.6	6.8	6.7	5.9	4.6	-
U3 - cycle-1 oracle	4.5	6.7	6.7	5.8	5.3	-
Mean IPDE administrations						
U2 - shared alpha	0.7	3.6	2.1	2.2	1.6	-
U2 - dose-specific alpha	0.7	3.5	2.1	2.3	1.6	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.7	3.6	2.1	2.2	1.6	-
U3 - dose-specific alpha	0.7	3.6	2.1	2.2	1.6	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1007
U2 - dose-specific alpha						1006.6
U2 - cycle-1 oracle						1603.3
U3 - shared alpha						1006.9
U3 - dose-specific alpha						1006.5
U3 - cycle-1 oracle						1603.3
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
MTD selection (%)						
U2 - shared alpha	61.8	13.5	4.2	1.1	0.2	19.2
U2 - dose-specific alpha	62.2	12.9	4.7	0.9	0.2	19.1
U2 - cycle-1 oracle	56.5	17.8	8.4	3.8	0.8	12.7
U3 - shared alpha	61.9	12.9	4.4	1.2	0.2	19.4
U3 - dose-specific alpha	62.2	12.6	4.6	1.3	0.1	19.2
U3 - cycle-1 oracle	56.8	17.8	8.5	3.6	0.8	12.5
OBD selection (%)						
U2 - shared alpha	66.2	9.7	1.5	0.6	0.1	21.9
U2 - dose-specific alpha	66.1	9.5	2.2	0.3	0.1	21.8
U2 - cycle-1 oracle	65.4	13.1	3.8	1.1	0.5	16.1
U3 - shared alpha	68.3	8.5	1.4	0.2	0.1	21.5
U3 - dose-specific alpha	67.5	9	1.6	0.2	0	21.7
U3 - cycle-1 oracle	66.9	12.6	3.2	0.8	0.3	16.2
Mean administrations						
U2 - shared alpha	20.1	3.6	1.7	0.5	0.2	-
U2 - dose-specific alpha	20.1	3.7	1.6	0.6	0.2	-
U2 - cycle-1 oracle	19.2	4	2.5	1	0.4	-
U3 - shared alpha	20.1	3.6	1.7	0.5	0.2	-
U3 - dose-specific alpha	20.1	3.6	1.7	0.6	0.2	-
U3 - cycle-1 oracle	19.1	4	2.5	1	0.4	-
Mean IPDE administrations						
U2 - shared alpha	5.6	1.5	0.5	0.2	0	-
U2 - dose-specific alpha	5.6	1.5	0.5	0.2	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	5.6	1.5	0.5	0.2	0	-
U3 - dose-specific alpha	5.6	1.5	0.5	0.2	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1011.5
U2 - dose-specific alpha						1009.8
U2 - cycle-1 oracle						1504
U3 - shared alpha						1011.8
U3 - dose-specific alpha						1010.8
U3 - cycle-1 oracle						1502.5

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
MTD selection (%)						
U2 - shared alpha	28.7	19.8	24.7	18.2	7.5	1.1
U2 - dose-specific alpha	29	19.7	24.9	17.2	8	1.2
U2 - cycle-1 oracle	23.4	15.5	23.7	20.7	16	0.7
U3 - shared alpha	28.8	20.5	24	17.5	8.1	1.1
U3 - dose-specific alpha	28.3	20.5	24.8	17.9	7.4	1.1
U3 - cycle-1 oracle	23.3	15.2	23.6	20.8	16.4	0.7
OBD selection (%)						
U2 - shared alpha	28.2	39.2	18.3	7.8	1.6	4.9
U2 - dose-specific alpha	29.4	39.1	18.1	7.1	1.4	4.9
U2 - cycle-1 oracle	24.1	36.5	20.4	9.9	3.2	5.9
U3 - shared alpha	31	41.7	14.7	6.6	1.1	4.9
U3 - dose-specific alpha	29.2	44.1	14.7	5.9	1.2	4.9
U3 - cycle-1 oracle	25.3	40.2	18.3	8.4	1.9	5.9
Mean administrations						
U2 - shared alpha	11.7	7.5	5.3	3.1	1.6	-
U2 - dose-specific alpha	11.8	7.5	5.2	3.1	1.5	-
U2 - cycle-1 oracle	10.1	7.3	5.6	3.8	2.2	-
U3 - shared alpha	11.8	7.4	5.4	3	1.5	-
U3 - dose-specific alpha	11.8	7.5	5.3	3	1.6	-
U3 - cycle-1 oracle	10.1	7.2	5.6	3.8	2.3	-
Mean IPDE administrations						
U2 - shared alpha	3.4	3	1.3	0.9	0.4	-
U2 - dose-specific alpha	3.4	3	1.3	0.9	0.4	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.4	3	1.3	0.9	0.4	-
U3 - dose-specific alpha	3.4	3	1.3	0.9	0.4	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1108.5
U2 - dose-specific alpha						1108.9
U2 - cycle-1 oracle						1608.4
U3 - shared alpha						1108.7
U3 - dose-specific alpha						1109.3
U3 - cycle-1 oracle						1608.4
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
MTD selection (%)						
U2 - shared alpha	65.7	7.5	1.9	0.2	0	24.7
U2 - dose-specific alpha	65.4	7.2	2.2	0.2	0	25
U2 - cycle-1 oracle	66.6	11.5	5.7	1.7	0.4	14.1
U3 - shared alpha	65.8	7	2.2	0.2	0	24.8
U3 - dose-specific alpha	65.2	7.5	2	0.2	0	25.1
U3 - cycle-1 oracle	66.8	11.5	5.4	1.9	0.4	14
OBD selection (%)						
U2 - shared alpha	27.5	5.4	0.9	0.1	0	66.1
U2 - dose-specific alpha	27.8	4.9	1.1	0.1	0	66.1
U2 - cycle-1 oracle	26	8.8	2.5	0.8	0.3	61.6
U3 - shared alpha	28.8	4.6	0.7	0.1	0	65.8
U3 - dose-specific alpha	28.6	4.9	0.5	0.1	0	65.9
U3 - cycle-1 oracle	27.6	7.7	1.8	1.1	0.2	61.6
Mean administrations						
U2 - shared alpha	14.6	2.8	1.1	0.2	0.1	-
U2 - dose-specific alpha	14.6	2.7	1.1	0.2	0.1	-
U2 - cycle-1 oracle	13.4	3.6	1.9	0.7	0.2	-
U3 - shared alpha	14.6	2.8	1.1	0.3	0.1	-
U3 - dose-specific alpha	14.7	2.8	1.1	0.2	0.1	-
U3 - cycle-1 oracle	13.4	3.6	1.9	0.8	0.2	-
Mean IPDE administrations						
U2 - shared alpha	4.4	1.4	0.4	0.1	0	-
U2 - dose-specific alpha	4.4	1.4	0.4	0.1	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	4.4	1.4	0.4	0.1	0	-
U3 - dose-specific alpha	4.4	1.4	0.4	0.1	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						686.2
U2 - dose-specific alpha						686
U2 - cycle-1 oracle						1095.9
U3 - shared alpha						686.3
U3 - dose-specific alpha						687.2
U3 - cycle-1 oracle						1095.3

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
MTD selection (%)						
U2 - shared alpha	28.4	15.9	17.9	17.6	19.4	0.8
U2 - dose-specific alpha	28.5	15.3	18.4	18.2	18.8	0.8
U2 - cycle-1 oracle	23.1	9.1	17.2	18	32.1	0.5
U3 - shared alpha	28.6	15.5	18.1	17.9	19.1	0.8
U3 - dose-specific alpha	28.3	15.7	18.5	17.4	19.3	0.8
U3 - cycle-1 oracle	23	8.8	17.6	17.8	32.3	0.5
OBD selection (%)						
U2 - shared alpha	11.2	13.7	15.6	11.1	15.7	32.7
U2 - dose-specific alpha	12	13.4	15.3	11.5	15.4	32.4
U2 - cycle-1 oracle	8.7	8.6	19.4	13.1	23.9	26.3
U3 - shared alpha	15.3	12.7	14.4	10.3	14.4	32.9
U3 - dose-specific alpha	16.1	12.6	13.7	10.7	13.8	33.1
U3 - cycle-1 oracle	13.8	11.1	16	11.9	20.8	26.4
Mean administrations						
U2 - shared alpha	8	4.5	5.2	3.4	3.5	-
U2 - dose-specific alpha	8	4.5	5.2	3.5	3.5	-
U2 - cycle-1 oracle	6.9	4.3	5.6	4	4.8	-
U3 - shared alpha	8	4.5	5.2	3.4	3.5	-
U3 - dose-specific alpha	7.9	4.5	5.3	3.5	3.4	-
U3 - cycle-1 oracle	6.9	4.3	5.5	4.1	4.8	-
Mean IPDE administrations						
U2 - shared alpha	2.2	2.6	1.8	1.4	1.1	-
U2 - dose-specific alpha	2.1	2.6	1.9	1.4	1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.1	2.6	1.9	1.4	1.1	-
U3 - dose-specific alpha	2.1	2.6	1.9	1.4	1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						859.5
U2 - dose-specific alpha						859.2
U2 - cycle-1 oracle						1424
U3 - shared alpha						859.4
U3 - dose-specific alpha						858.5
U3 - cycle-1 oracle						1424.4
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
MTD selection (%)						
U2 - shared alpha	56.1	20.7	12.4	4.9	0.9	5
U2 - dose-specific alpha	56.5	20.4	12.4	4.6	1	5.1
U2 - cycle-1 oracle	47.9	18.8	15.9	9.7	4.4	3.3
U3 - shared alpha	56.6	19.8	12.7	4.6	1.1	5.2
U3 - dose-specific alpha	56.3	20.2	12.7	4.5	1.1	5.2
U3 - cycle-1 oracle	47.8	19	15.6	9.7	4.6	3.3
OBD selection (%)						
U2 - shared alpha	8.8	25	9.4	2.2	0.6	54
U2 - dose-specific alpha	8.8	25.7	8.3	1.9	0.7	54.6
U2 - cycle-1 oracle	6.7	29.1	11.5	3.9	1.9	46.9
U3 - shared alpha	9.4	26.1	8.2	1.4	0.6	54.3
U3 - dose-specific alpha	9.4	26.4	8	1.4	0.5	54.3
U3 - cycle-1 oracle	7.2	29.3	11.9	3.1	1.5	47
Mean administrations						
U2 - shared alpha	9.8	5.4	3.4	1.4	0.4	-
U2 - dose-specific alpha	9.8	5.4	3.4	1.4	0.4	-
U2 - cycle-1 oracle	8.4	6	4.1	2.1	1	-
U3 - shared alpha	9.8	5.5	3.3	1.4	0.4	-
U3 - dose-specific alpha	9.8	5.4	3.4	1.4	0.4	-
U3 - cycle-1 oracle	8.4	5.9	4.1	2.1	1	-
Mean IPDE administrations						
U2 - shared alpha	2.7	2.4	0.9	0.4	0.1	-
U2 - dose-specific alpha	2.7	2.4	0.9	0.5	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.7	2.4	0.9	0.4	0.1	-
U3 - dose-specific alpha	2.7	2.4	0.9	0.4	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						750.6
U2 - dose-specific alpha						748.4
U2 - cycle-1 oracle						1196.2
U3 - shared alpha						748.5
U3 - dose-specific alpha						748.5
U3 - cycle-1 oracle						1193.5

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
MTD selection (%)						
U2 - shared alpha	61.3	27.1	6.5	0	0	5.1
U2 - dose-specific alpha	60.5	27.7	6.5	0	0	5.3
U2 - cycle-1 oracle	48.8	32.6	12.8	2.2	0	3.6
U3 - shared alpha	61	27.5	6.4	0	0	5.1
U3 - dose-specific alpha	61.3	26.9	6.8	0	0	5
U3 - cycle-1 oracle	49.3	32.2	12.7	2.2	0	3.6
OBD selection (%)						
U2 - shared alpha	67.9	16.6	1.2	0	0	14.3
U2 - dose-specific alpha	66.6	17.4	1.4	0	0	14.6
U2 - cycle-1 oracle	58.8	20.8	3.7	0.8	0	15.9
U3 - shared alpha	69	15.8	0.9	0	0	14.3
U3 - dose-specific alpha	69.1	15.9	0.9	0	0	14.1
U3 - cycle-1 oracle	61.4	19.5	2.4	0.6	0	16.1
Mean administrations						
U2 - shared alpha	19.1	5.5	2.4	0.5	0	-
U2 - dose-specific alpha	19.2	5.4	2.4	0.5	0	-
U2 - cycle-1 oracle	16.9	5.9	3.5	1	0.1	-
U3 - shared alpha	19.1	5.5	2.4	0.5	0	-
U3 - dose-specific alpha	19.2	5.4	2.4	0.5	0	-
U3 - cycle-1 oracle	17	5.9	3.5	1	0.1	-
Mean IPDE administrations						
U2 - shared alpha	6	2.4	0.8	0.2	0	-
U2 - dose-specific alpha	6	2.5	0.8	0.2	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	6	2.5	0.8	0.2	0	-
U3 - dose-specific alpha	6	2.5	0.8	0.2	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						994.2
U2 - dose-specific alpha						994.1
U2 - cycle-1 oracle						1520.9
U3 - shared alpha						994.8
U3 - dose-specific alpha						994.6
U3 - cycle-1 oracle						1520.8
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
MTD selection (%)						
U2 - shared alpha	4.7	3.8	30	44.9	16.5	0.1
U2 - dose-specific alpha	4.7	3.6	30.1	44.4	17.1	0.1
U2 - cycle-1 oracle	4	2.3	21	43.5	29.1	0.1
U3 - shared alpha	4.6	3.4	31.2	44.2	16.5	0.1
U3 - dose-specific alpha	4.7	3.3	29.9	45.5	16.5	0.1
U3 - cycle-1 oracle	4	2.5	20.5	43.8	29.1	0.1
OBD selection (%)						
U2 - shared alpha	14.8	17.1	12.6	37.1	12.9	5.5
U2 - dose-specific alpha	15	18	11.2	37.1	12.9	5.8
U2 - cycle-1 oracle	13.8	12.5	8.7	37.1	22.4	5.5
U3 - shared alpha	21.1	17.6	10	33.4	12.1	5.8
U3 - dose-specific alpha	22.2	16.3	8.7	35.4	11.9	5.5
U3 - cycle-1 oracle	19.1	12.9	7.6	35	20	5.4
Mean administrations						
U2 - shared alpha	4.9	5.3	6	7.4	5.5	-
U2 - dose-specific alpha	5	5.2	6	7.4	5.5	-
U2 - cycle-1 oracle	4.6	4.8	5.4	7.6	6.8	-
U3 - shared alpha	5	5.3	6	7.3	5.5	-
U3 - dose-specific alpha	4.9	5.3	6	7.4	5.5	-
U3 - cycle-1 oracle	4.7	4.8	5.4	7.5	6.8	-
Mean IPDE administrations						
U2 - shared alpha	0.9	3.2	2	2.8	1.3	-
U2 - dose-specific alpha	0.9	3.2	1.9	2.8	1.3	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.9	3.2	1.9	2.7	1.3	-
U3 - dose-specific alpha	0.9	3.2	1.9	2.8	1.3	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1047.3
U2 - dose-specific alpha						1047.4
U2 - cycle-1 oracle						1614.1
U3 - shared alpha						1046.8
U3 - dose-specific alpha						1047.4
U3 - cycle-1 oracle						1614.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
MTD selection (%)						
U2 - shared alpha	54.2	21.7	14.3	5.2	0.9	3.7
U2 - dose-specific alpha	53.7	22.3	14.8	4.6	0.8	3.8
U2 - cycle-1 oracle	43.8	20.1	19.2	10.2	4.5	2.2
U3 - shared alpha	53.9	22.3	14.6	4.8	0.7	3.7
U3 - dose-specific alpha	54.1	21.3	15	5.1	0.8	3.7
U3 - cycle-1 oracle	44.2	19.8	18.8	11	4	2.2
OBD selection (%)						
U2 - shared alpha	11.4	26.2	9.7	1.9	0.6	50.2
U2 - dose-specific alpha	10.6	27	9.6	1.9	0.6	50.3
U2 - cycle-1 oracle	8.4	30.2	13.3	4.2	2	41.9
U3 - shared alpha	12.1	26.6	9.2	1.4	0.5	50.2
U3 - dose-specific alpha	11.8	26	9.4	1.8	0.6	50.4
U3 - cycle-1 oracle	9.6	31.4	11.4	4	1.4	42.2
Mean administrations						
U2 - shared alpha	9.6	5.8	3.7	1.4	0.5	-
U2 - dose-specific alpha	9.6	5.8	3.6	1.5	0.5	-
U2 - cycle-1 oracle	8.2	6.4	4.5	2.4	1	-
U3 - shared alpha	9.6	5.8	3.7	1.4	0.5	-
U3 - dose-specific alpha	9.6	5.8	3.7	1.4	0.5	-
U3 - cycle-1 oracle	8.2	6.3	4.5	2.4	1.1	-
Mean IPDE administrations						
U2 - shared alpha	2.7	2.7	1.1	0.6	0.2	-
U2 - dose-specific alpha	2.7	2.8	1.1	0.6	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.7	2.7	1.1	0.5	0.2	-
U3 - dose-specific alpha	2.7	2.7	1.1	0.5	0.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						757.5
U2 - dose-specific alpha						756.8
U2 - cycle-1 oracle						1240.4
U3 - shared alpha						756.7
U3 - dose-specific alpha						755.9
U3 - cycle-1 oracle						1239.8
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
MTD selection (%)						
U2 - shared alpha	13.1	11.5	30.2	29.5	15.5	0.2
U2 - dose-specific alpha	13	11.6	29.9	29.7	15.6	0.2
U2 - cycle-1 oracle	9.5	6.9	24.8	31	27.6	0.2
U3 - shared alpha	13.2	12.3	28.6	29.9	15.8	0.2
U3 - dose-specific alpha	13.2	11.4	29.7	29.7	15.8	0.2
U3 - cycle-1 oracle	9.4	7	25	31.7	26.7	0.2
OBD selection (%)						
U2 - shared alpha	18	18.5	14	24.4	8.7	16.4
U2 - dose-specific alpha	18.7	16.2	16.5	22.8	9.3	16.5
U2 - cycle-1 oracle	16.2	12.5	15.1	28.2	14.9	13.1
U3 - shared alpha	24.8	16.8	11.7	21.8	7.8	17.1
U3 - dose-specific alpha	27.2	16.7	10.9	21	7.5	16.7
U3 - cycle-1 oracle	25.4	15.1	11.3	24.6	10.6	13
Mean administrations						
U2 - shared alpha	6.7	5.7	6.1	5.2	3.7	-
U2 - dose-specific alpha	6.7	5.5	6	5.3	3.9	-
U2 - cycle-1 oracle	5.9	5.3	6	6.2	4.7	-
U3 - shared alpha	6.6	5.7	6	5.3	3.8	-
U3 - dose-specific alpha	6.7	5.7	6	5.3	3.8	-
U3 - cycle-1 oracle	6	5.3	6	6.2	4.7	-
Mean IPDE administrations						
U2 - shared alpha	1.6	3.3	2.1	2	1.2	-
U2 - dose-specific alpha	1.7	3.2	2.1	2.1	1.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.6	3.2	2.1	2.1	1.2	-
U3 - dose-specific alpha	1.6	3.2	2.1	2.1	1.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						947.5
U2 - dose-specific alpha						948.3
U2 - cycle-1 oracle						1556.1
U3 - shared alpha						947.9
U3 - dose-specific alpha						945.9
U3 - cycle-1 oracle						1557.1

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
MTD selection (%)						
U2 - shared alpha	23.9	30.9	30.9	11.5	2.3	0.5
U2 - dose-specific alpha	23.7	30.3	32.1	11.2	2.2	0.5
U2 - cycle-1 oracle	18.6	20	35	17	9	0.4
U3 - shared alpha	24.2	29.6	32.6	10.5	2.6	0.5
U3 - dose-specific alpha	24	28.7	32.7	12.1	2	0.5
U3 - cycle-1 oracle	18.7	20.3	34.9	17.3	8.4	0.4
OBD selection (%)						
U2 - shared alpha	41.7	26.8	9.2	2.9	0.2	19.2
U2 - dose-specific alpha	42.9	26.2	9.3	2.7	0.1	18.8
U2 - cycle-1 oracle	40.4	24.1	11.8	3.7	1.1	18.9
U3 - shared alpha	50.2	24.2	5.1	1.1	0.2	19.2
U3 - dose-specific alpha	49.1	25.8	4.8	1.1	0.1	19.1
U3 - cycle-1 oracle	49.6	22.2	7.1	2.1	0.4	18.6
Mean administrations						
U2 - shared alpha	11.3	7.4	5.4	2.5	0.8	-
U2 - dose-specific alpha	11.3	7.3	5.4	2.4	0.9	-
U2 - cycle-1 oracle	10.1	6.7	5.9	3.2	1.5	-
U3 - shared alpha	11.2	7.5	5.4	2.4	0.9	-
U3 - dose-specific alpha	11.3	7.4	5.4	2.4	0.8	-
U3 - cycle-1 oracle	10.2	6.6	5.9	3.2	1.5	-
Mean IPDE administrations						
U2 - shared alpha	3.5	3.7	1.9	1.1	0.3	-
U2 - dose-specific alpha	3.5	3.7	1.9	1.1	0.4	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.4	3.8	1.9	1.1	0.4	-
U3 - dose-specific alpha	3.5	3.7	1.9	1.1	0.3	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						923.1
U2 - dose-specific alpha						923.5
U2 - cycle-1 oracle						1523.1
U3 - shared alpha						922.6
U3 - dose-specific alpha						923.2
U3 - cycle-1 oracle						1523.4
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
MTD selection (%)						
U2 - shared alpha	0.7	0.7	3	13.3	82.3	0
U2 - dose-specific alpha	0.7	0.7	3.1	12.3	83.2	0
U2 - cycle-1 oracle	0.7	0.6	2.7	10.9	85.1	0
U3 - shared alpha	0.7	0.7	3.2	12.9	82.5	0
U3 - dose-specific alpha	0.7	0.7	3.1	11.9	83.6	0
U3 - cycle-1 oracle	0.7	0.6	2.7	10.5	85.5	0
OBD selection (%)						
U2 - shared alpha	27.8	28.1	17.3	15.2	11.5	0.1
U2 - dose-specific alpha	27	27.2	16.6	17	12.1	0.1
U2 - cycle-1 oracle	30.4	26.3	16	16	11.2	0.1
U3 - shared alpha	34.5	31.6	15	11.9	6.9	0.1
U3 - dose-specific alpha	34.1	32.7	14.1	12	7	0.1
U3 - cycle-1 oracle	37.9	24.6	16.8	13.5	7.1	0.1
Mean administrations						
U2 - shared alpha	5.8	5.9	5.5	5.7	7	-
U2 - dose-specific alpha	5.7	5.9	5.5	5.8	7	-
U2 - cycle-1 oracle	6	5.6	5.4	6	7	-
U3 - shared alpha	5.8	5.9	5.5	5.8	7	-
U3 - dose-specific alpha	5.7	5.9	5.5	5.8	7	-
U3 - cycle-1 oracle	5.9	5.6	5.5	6	7	-
Mean IPDE administrations						
U2 - shared alpha	1.2	3.3	1.9	2.6	2.7	-
U2 - dose-specific alpha	1.1	3.3	1.9	2.6	2.7	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.1	3.3	1.9	2.7	2.7	-
U3 - dose-specific alpha	1.1	3.3	1.9	2.6	2.7	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1007.2
U2 - dose-specific alpha						1006.6
U2 - cycle-1 oracle						1662.9
U3 - shared alpha						1006.8
U3 - dose-specific alpha						1006.8
U3 - cycle-1 oracle						1662.9

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
MTD selection (%)						
U2 - shared alpha	20.5	46.3	30	3	0	0.2
U2 - dose-specific alpha	21.4	46	29.6	2.8	0	0.2
U2 - cycle-1 oracle	15.7	41.7	35.4	6.8	0.2	0.2
U3 - shared alpha	20.6	45.4	30.9	2.9	0	0.2
U3 - dose-specific alpha	20.8	46.4	29.7	2.9	0	0.2
U3 - cycle-1 oracle	16	40.5	36.4	6.7	0.2	0.2
OBD selection (%)						
U2 - shared alpha	31.5	53.9	13.1	1	0	0.5
U2 - dose-specific alpha	32.8	53.9	11.9	0.8	0	0.6
U2 - cycle-1 oracle	30	50.5	16.9	1.6	0	1
U3 - shared alpha	34.6	53.9	10.3	0.7	0	0.5
U3 - dose-specific alpha	35.2	53.6	10.1	0.6	0	0.5
U3 - cycle-1 oracle	33.1	51.1	13.5	1.2	0.1	1
Mean administrations						
U2 - shared alpha	11.1	10.5	6.5	1.5	0.2	-
U2 - dose-specific alpha	11	10.6	6.5	1.5	0.2	-
U2 - cycle-1 oracle	10	10	7.2	2.3	0.4	-
U3 - shared alpha	11.1	10.5	6.5	1.6	0.2	-
U3 - dose-specific alpha	11.1	10.5	6.5	1.5	0.2	-
U3 - cycle-1 oracle	10	10	7.2	2.3	0.4	-
Mean IPDE administrations						
U2 - shared alpha	2.9	3.6	1.5	0.5	0.1	-
U2 - dose-specific alpha	2.8	3.6	1.5	0.4	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.9	3.6	1.5	0.5	0.1	-
U3 - dose-specific alpha	2.9	3.6	1.5	0.5	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1183.5
U2 - dose-specific alpha						1183.9
U2 - cycle-1 oracle						1653.1
U3 - shared alpha						1183.1
U3 - dose-specific alpha						1183.5
U3 - cycle-1 oracle						1653.1
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
MTD selection (%)						
U2 - shared alpha	0.7	0.2	6	34.3	58.8	0
U2 - dose-specific alpha	0.7	0.3	5.7	34	59.3	0
U2 - cycle-1 oracle	0.7	0.1	3.7	24.9	70.6	0
U3 - shared alpha	0.7	0.2	5.7	34.4	59	0
U3 - dose-specific alpha	0.7	0.3	5.7	34.7	58.6	0
U3 - cycle-1 oracle	0.7	0.1	3.9	25	70.3	0
OBD selection (%)						
U2 - shared alpha	9.7	11.9	11.5	17.9	48	1
U2 - dose-specific alpha	9.5	12.4	11.2	17.5	48.5	0.9
U2 - cycle-1 oracle	10.3	8.1	9.9	14.9	55.9	0.9
U3 - shared alpha	14.5	14.3	11.2	16.1	43	0.9
U3 - dose-specific alpha	15.4	15.2	11	15.4	42	1
U3 - cycle-1 oracle	16.9	12.3	9.4	12.7	47.7	1
Mean administrations						
U2 - shared alpha	3.7	4	4.9	5.7	11.5	-
U2 - dose-specific alpha	3.7	4	4.9	5.7	11.6	-
U2 - cycle-1 oracle	3.7	4	4.8	5.8	11.6	-
U3 - shared alpha	3.7	4	4.9	5.7	11.6	-
U3 - dose-specific alpha	3.7	4	4.8	5.8	11.6	-
U3 - cycle-1 oracle	3.7	4	4.8	5.8	11.6	-
Mean IPDE administrations						
U2 - shared alpha	0.3	2.8	1.6	2.6	3.2	-
U2 - dose-specific alpha	0.3	2.8	1.6	2.5	3.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.3	2.8	1.6	2.6	3.2	-
U3 - dose-specific alpha	0.4	2.8	1.6	2.6	3.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1065.9
U2 - dose-specific alpha						1065.6
U2 - cycle-1 oracle						1655.7
U3 - shared alpha						1066.6
U3 - dose-specific alpha						1065.1
U3 - cycle-1 oracle						1655.3

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
MTD selection (%)						
U2 - shared alpha	30.7	46.1	22	1	0	0.2
U2 - dose-specific alpha	31	46.6	21.1	1.1	0	0.2
U2 - cycle-1 oracle	22	41.7	31	5	0.1	0.2
U3 - shared alpha	30.3	47.8	20.8	0.9	0	0.2
U3 - dose-specific alpha	30.8	46.9	21	1.1	0	0.2
U3 - cycle-1 oracle	22.3	42	30.8	4.6	0.1	0.2
OBD selection (%)						
U2 - shared alpha	22.2	48.8	11.2	0.5	0	17.3
U2 - dose-specific alpha	22	49	10.7	0.7	0	17.6
U2 - cycle-1 oracle	15.5	51.2	16.1	1.4	0.1	15.7
U3 - shared alpha	24.8	48.2	9.4	0.4	0	17.2
U3 - dose-specific alpha	23.8	48.6	9.1	0.7	0	17.8
U3 - cycle-1 oracle	18.8	50.9	13.1	1.2	0	16
Mean administrations						
U2 - shared alpha	9	10.6	6	1.3	0.2	-
U2 - dose-specific alpha	9	10.5	6	1.3	0.2	-
U2 - cycle-1 oracle	7.5	10.5	6.9	2.1	0.3	-
U3 - shared alpha	9	10.6	6	1.3	0.2	-
U3 - dose-specific alpha	9	10.6	6.1	1.3	0.2	-
U3 - cycle-1 oracle	7.6	10.4	6.9	2.1	0.3	-
Mean IPDE administrations						
U2 - shared alpha	2.5	4.6	1.7	0.5	0	-
U2 - dose-specific alpha	2.5	4.6	1.8	0.5	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.5	4.6	1.7	0.5	0	-
U3 - dose-specific alpha	2.5	4.5	1.8	0.5	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						974.7
U2 - dose-specific alpha						976.3
U2 - cycle-1 oracle						1514.3
U3 - shared alpha						976.4
U3 - dose-specific alpha						976.1
U3 - cycle-1 oracle						1513.5
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
MTD selection (%)						
U2 - shared alpha	8.1	22.3	48.3	19.4	1.8	0.1
U2 - dose-specific alpha	8.3	21.7	48.9	18.5	2.5	0.1
U2 - cycle-1 oracle	6.3	14.7	44	25.5	9.4	0.1
U3 - shared alpha	8.3	20.9	49.9	18.3	2.5	0.1
U3 - dose-specific alpha	8.6	21.3	49.6	17.5	2.9	0.1
U3 - cycle-1 oracle	6.4	13.6	45.3	25.2	9.4	0.1
OBD selection (%)						
U2 - shared alpha	10.6	48.9	26.6	6.1	0.1	7.7
U2 - dose-specific alpha	10.6	47.7	27.7	5.9	0.3	7.8
U2 - cycle-1 oracle	11.4	46.2	26.8	8.3	1.1	6.2
U3 - shared alpha	14.8	49.7	22.7	4.7	0.4	7.7
U3 - dose-specific alpha	14.3	51.2	22.3	3.9	0.4	7.9
U3 - cycle-1 oracle	16.6	47.8	22.4	6.3	0.8	6.1
Mean administrations						
U2 - shared alpha	5.5	9.5	8.5	3.7	1.3	-
U2 - dose-specific alpha	5.4	9.5	8.5	3.7	1.4	-
U2 - cycle-1 oracle	5.1	9	8.3	4.5	2	-
U3 - shared alpha	5.5	9.5	8.4	3.7	1.4	-
U3 - dose-specific alpha	5.4	9.5	8.5	3.7	1.4	-
U3 - cycle-1 oracle	5	8.9	8.5	4.5	2	-
Mean IPDE administrations						
U2 - shared alpha	1.1	4.6	2.6	1.4	0.5	-
U2 - dose-specific alpha	1.1	4.6	2.6	1.4	0.5	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.1	4.6	2.6	1.4	0.5	-
U3 - dose-specific alpha	1.1	4.6	2.6	1.4	0.5	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1004.6
U2 - dose-specific alpha						1004.8
U2 - cycle-1 oracle						1599.2
U3 - shared alpha						1004.9
U3 - dose-specific alpha						1004.6
U3 - cycle-1 oracle						1598.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
MTD selection (%)						
U2 - shared alpha	49.1	0.8	0	0	0	50.1
U2 - dose-specific alpha	48.8	0.9	0	0	0	50.3
U2 - cycle-1 oracle	59.6	3.1	0.1	0	0	37.2
U3 - shared alpha	49.8	0.8	0	0	0	49.4
U3 - dose-specific alpha	49.7	0.9	0	0	0	49.4
U3 - cycle-1 oracle	59.6	3	0.1	0	0	37.3
OBD selection (%)						
U2 - shared alpha	47.6	0.4	0	0	0	52
U2 - dose-specific alpha	47.3	0.5	0	0	0	52.2
U2 - cycle-1 oracle	59.8	1.6	0	0	0	38.6
U3 - shared alpha	47.3	0.4	0	0	0	52.3
U3 - dose-specific alpha	47.2	0.5	0	0	0	52.3
U3 - cycle-1 oracle	60.1	1.3	0	0	0	38.6
Mean administrations						
U2 - shared alpha	19.7	1.6	0.3	0	0	-
U2 - dose-specific alpha	19.7	1.6	0.3	0	0	-
U2 - cycle-1 oracle	20.9	2.1	0.6	0	0	-
U3 - shared alpha	19.7	1.6	0.3	0	0	-
U3 - dose-specific alpha	19.7	1.6	0.3	0	0	-
U3 - cycle-1 oracle	20.9	2	0.6	0	0	-
Mean IPDE administrations						
U2 - shared alpha	4	0.5	0.1	0	0	-
U2 - dose-specific alpha	4	0.5	0.1	0	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	4	0.5	0.1	0	0	-
U3 - dose-specific alpha	4	0.5	0.1	0	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						933.8
U2 - dose-specific alpha						935
U2 - cycle-1 oracle						1307.9
U3 - shared alpha						934.4
U3 - dose-specific alpha						933.6
U3 - cycle-1 oracle						1308.6

Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
MTD selection (%)						
U2 - shared alpha	25.7	53.9	19.7	0.6	0	0.1
U2 - dose-specific alpha	25.8	54.6	18.8	0.7	0	0.1
U2 - cycle-1 oracle	20	49.1	28.6	2.2	0	0.1
U3 - shared alpha	25.8	54.8	18.7	0.6	0	0.1
U3 - dose-specific alpha	25.2	55.6	18.2	0.9	0	0.1
U3 - cycle-1 oracle	20	49.7	28.4	1.8	0	0.1
OBD selection (%)						
U2 - shared alpha	26.4	53	8.9	0.5	0	11.2
U2 - dose-specific alpha	27.4	53.3	7.8	0.5	0	11
U2 - cycle-1 oracle	23.4	53.4	10.8	1.4	0	11
U3 - shared alpha	29.6	52.3	6.9	0.4	0	10.8
U3 - dose-specific alpha	28.4	53.6	6.3	0.6	0	11.1
U3 - cycle-1 oracle	26.5	51.7	9.2	1.3	0	11.3
Mean administrations						
U2 - shared alpha	8.8	13	5.2	1	0.1	-
U2 - dose-specific alpha	8.8	13	5.2	1	0.1	-
U2 - cycle-1 oracle	7.7	12.2	6.3	1.6	0.2	-
U3 - shared alpha	8.7	13	5.3	1	0.1	-
U3 - dose-specific alpha	8.8	13	5.2	1	0.1	-
U3 - cycle-1 oracle	7.8	12	6.3	1.7	0.2	-
Mean IPDE administrations						
U2 - shared alpha	2.3	5.6	1.6	0.4	0	-
U2 - dose-specific alpha	2.3	5.6	1.5	0.4	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.3	5.6	1.5	0.4	0	-
U3 - dose-specific alpha	2.3	5.6	1.5	0.4	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						994.8
U2 - dose-specific alpha						994.1
U2 - cycle-1 oracle						1555.7
U3 - shared alpha						995.4
U3 - dose-specific alpha						992.5
U3 - cycle-1 oracle						1552.8

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
MTD selection (%)						
U2 - shared alpha	75.3	2.4	0	0	0	22.3
U2 - dose-specific alpha	74.9	2.4	0	0	0	22.7
U2 - cycle-1 oracle	80.7	5.1	0.4	0	0	13.8
U3 - shared alpha	75.2	2.4	0	0	0	22.4
U3 - dose-specific alpha	74.6	2.3	0	0	0	23.1
U3 - cycle-1 oracle	81	4.7	0.4	0	0	13.9
OBD selection (%)						
U2 - shared alpha	6.3	1.6	0	0	0	92.1
U2 - dose-specific alpha	6.1	1.6	0	0	0	92.3
U2 - cycle-1 oracle	7	2.5	0.3	0	0	90.2
U3 - shared alpha	6.3	1.6	0	0	0	92.1
U3 - dose-specific alpha	6.1	1.5	0	0	0	92.4
U3 - cycle-1 oracle	7	2.4	0.3	0	0	90.3
Mean administrations						
U2 - shared alpha	11.8	1.7	0.2	0	0	-
U2 - dose-specific alpha	11.7	1.7	0.2	0	0	-
U2 - cycle-1 oracle	11.7	2.5	0.6	0	0	-
U3 - shared alpha	11.8	1.7	0.2	0	0	-
U3 - dose-specific alpha	11.7	1.7	0.2	0	0	-
U3 - cycle-1 oracle	11.7	2.5	0.6	0	0	-
Mean IPDE administrations						
U2 - shared alpha	3.4	1	0.1	0	0	-
U2 - dose-specific alpha	3.4	1	0.1	0	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.4	1	0.1	0	0	-
U3 - dose-specific alpha	3.4	1	0.1	0	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - shared alpha						498.7
U2 - dose-specific alpha						498.4
U2 - cycle-1 oracle						813.5
U3 - shared alpha						498.3
U3 - dose-specific alpha						497.9
U3 - cycle-1 oracle						814.4
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
MTD selection (%)						
U2 - shared alpha	0	0.1	0.6	5.4	93.9	0
U2 - dose-specific alpha	0	0.1	0.6	5.8	93.5	0
U2 - cycle-1 oracle	0	0.1	0.7	3.8	95.4	0
U3 - shared alpha	0	0.1	0.6	5.2	94.1	0
U3 - dose-specific alpha	0	0.1	0.6	5.7	93.6	0
U3 - cycle-1 oracle	0	0.1	0.7	4.1	95.1	0
OBD selection (%)						
U2 - shared alpha	0.6	1.7	4.3	47.6	45.7	0.1
U2 - dose-specific alpha	0.7	1.7	4	49.9	43.6	0.1
U2 - cycle-1 oracle	1.4	1.2	5.3	46.2	45.8	0.1
U3 - shared alpha	1.7	2.4	5	50.5	40.3	0.1
U3 - dose-specific alpha	1.8	2.6	4.8	51.1	39.6	0.1
U3 - cycle-1 oracle	2.2	2.4	8	49.6	37.7	0.1
Mean administrations						
U2 - shared alpha	3.2	3.1	3.6	8.3	11.8	-
U2 - dose-specific alpha	3.2	3.1	3.6	8.4	11.7	-
U2 - cycle-1 oracle	3.3	3.2	3.8	8.6	11.1	-
U3 - shared alpha	3.2	3.1	3.6	8.3	11.7	-
U3 - dose-specific alpha	3.2	3.1	3.6	8.5	11.5	-
U3 - cycle-1 oracle	3.3	3.3	3.8	8.6	11.1	-
Mean IPDE administrations						
U2 - shared alpha	0.1	2.4	1.1	3.2	3.2	-
U2 - dose-specific alpha	0.1	2.4	1.1	3.2	3.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.1	2.4	1.1	3.2	3.2	-
U3 - dose-specific alpha	0.1	2.4	1.1	3.3	3.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - shared alpha						1101.3
U2 - dose-specific alpha						1101.3
U2 - cycle-1 oracle						1663.8
U3 - shared alpha						1101.1
U3 - dose-specific alpha						1101
U3 - cycle-1 oracle						1663.5

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
MTD selection (%)						
U2 - shared alpha	14.4	40.3	40.3	4.8	0	0.2
U2 - dose-specific alpha	14	40.9	40.1	4.8	0	0.2
U2 - cycle-1 oracle	11.4	34.2	45.2	8.7	0.3	0.2
U3 - shared alpha	14	41.4	39.4	5	0	0.2
U3 - dose-specific alpha	14.2	40.3	40.8	4.5	0	0.2
U3 - cycle-1 oracle	11.4	33.2	46	8.8	0.4	0.2
OBD selection (%)						
U2 - shared alpha	36.2	33.5	24.9	3.3	0	2.1
U2 - dose-specific alpha	35.4	35.2	24.2	3.2	0	2
U2 - cycle-1 oracle	32.5	30.4	30.3	4.1	0.1	2.6
U3 - shared alpha	40.4	34.2	20.7	2.7	0	2
U3 - dose-specific alpha	40.6	33.6	21.7	2	0	2.1
U3 - cycle-1 oracle	37.6	29.9	26.5	3.4	0	2.6
Mean administrations						
U2 - shared alpha	10.1	8.7	8.6	1.9	0.3	-
U2 - dose-specific alpha	10.1	8.8	8.5	2	0.2	-
U2 - cycle-1 oracle	9	8.3	9.3	2.6	0.4	-
U3 - shared alpha	10.1	8.8	8.6	1.9	0.2	-
U3 - dose-specific alpha	10.1	8.8	8.5	1.9	0.2	-
U3 - cycle-1 oracle	9.1	8.2	9.3	2.6	0.4	-
Mean IPDE administrations						
U2 - shared alpha	2.9	3.8	2.4	0.7	0.1	-
U2 - dose-specific alpha	2.8	3.9	2.4	0.7	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.8	3.8	2.4	0.7	0.1	-
U3 - dose-specific alpha	2.8	3.9	2.3	0.7	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1095.2
U2 - dose-specific alpha						1095.1
U2 - cycle-1 oracle						1643.7
U3 - shared alpha						1094.7
U3 - dose-specific alpha						1094.5
U3 - cycle-1 oracle						1643.8

Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
MTD selection (%)						
U2 - shared alpha	32.2	34.2	28.2	4.6	0	0.8
U2 - dose-specific alpha	31.7	35.1	27.8	4.6	0	0.8
U2 - cycle-1 oracle	25.9	28.5	34.5	10.5	0.1	0.5
U3 - shared alpha	32.1	35.6	27.3	4.2	0	0.8
U3 - dose-specific alpha	32.5	34.3	28.4	4	0	0.8
U3 - cycle-1 oracle	25.8	28.4	35.5	9.5	0.3	0.5
OBD selection (%)						
U2 - shared alpha	27.3	29.6	20.9	2.5	0	19.7
U2 - dose-specific alpha	27.9	29	20.5	2.7	0	19.9
U2 - cycle-1 oracle	21.6	25	26.2	5.7	0	21.5
U3 - shared alpha	30.3	29.3	18.2	2	0	20.2
U3 - dose-specific alpha	30.6	29.1	18.5	1.9	0	19.9
U3 - cycle-1 oracle	23.9	27.2	22.9	4.8	0	21.2
Mean administrations						
U2 - shared alpha	10.7	7.3	6.9	1.8	0.2	-
U2 - dose-specific alpha	10.8	7.3	6.8	1.9	0.2	-
U2 - cycle-1 oracle	9	6.5	8	2.8	0.4	-
U3 - shared alpha	10.8	7.3	6.8	1.8	0.3	-
U3 - dose-specific alpha	10.7	7.3	6.7	1.9	0.2	-
U3 - cycle-1 oracle	9	6.7	7.9	2.8	0.4	-
Mean IPDE administrations						
U2 - shared alpha	3.2	3.4	2	0.7	0	-
U2 - dose-specific alpha	3.2	3.4	2	0.7	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.2	3.4	2	0.7	0.1	-
U3 - dose-specific alpha	3.2	3.5	2	0.7	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						964.2
U2 - dose-specific alpha						964.7
U2 - cycle-1 oracle						1487.4
U3 - shared alpha						963.5
U3 - dose-specific alpha						964
U3 - cycle-1 oracle						1487.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
MTD selection (%)						
U2 - shared alpha	17.7	29.6	42.6	9.7	0.2	0.2
U2 - dose-specific alpha	17.7	29.8	42.2	9.9	0.2	0.2
U2 - cycle-1 oracle	13.8	22	46.7	15.6	1.7	0.2
U3 - shared alpha	17.8	29.6	42.5	9.7	0.2	0.2
U3 - dose-specific alpha	17.6	30.1	41.6	10.3	0.2	0.2
U3 - cycle-1 oracle	13.7	21.5	46.6	16.3	1.7	0.2
OBD selection (%)						
U2 - shared alpha	19.1	27.6	37.8	3.5	0	12
U2 - dose-specific alpha	18.9	28.7	36.3	4.2	0	11.9
U2 - cycle-1 oracle	16	22.5	41.7	7.3	0.9	11.6
U3 - shared alpha	23	28.7	33	3.4	0	11.9
U3 - dose-specific alpha	23.8	27.9	32.3	3.9	0	12.1
U3 - cycle-1 oracle	19.2	24.1	37.9	6.7	0.5	11.6
Mean administrations						
U2 - shared alpha	8.2	7.1	9	3.1	0.8	-
U2 - dose-specific alpha	8.2	7	9.1	3.1	0.8	-
U2 - cycle-1 oracle	7	6.6	9.5	3.9	1.2	-
U3 - shared alpha	8.2	7.1	9	3.1	0.8	-
U3 - dose-specific alpha	8.2	7	9.1	3.1	0.7	-
U3 - cycle-1 oracle	7	6.6	9.5	3.9	1.2	-
Mean IPDE administrations						
U2 - shared alpha	2.2	3.4	2.6	1	0.2	-
U2 - dose-specific alpha	2.2	3.4	2.6	1	0.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.2	3.4	2.6	1	0.2	-
U3 - dose-specific alpha	2.2	3.4	2.6	1	0.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1030
U2 - dose-specific alpha						1029.9
U2 - cycle-1 oracle						1565.2
U3 - shared alpha						1029.8
U3 - dose-specific alpha						1029.4
U3 - cycle-1 oracle						1564.8

Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
MTD selection (%)						
U2 - shared alpha	63.6	28.7	4.4	0	0	3.3
U2 - dose-specific alpha	63	29	4.7	0	0	3.3
U2 - cycle-1 oracle	53	31.7	12.2	0.9	0	2.2
U3 - shared alpha	63	29.2	4.4	0	0	3.4
U3 - dose-specific alpha	63.7	28.1	4.9	0	0	3.3
U3 - cycle-1 oracle	52.8	32.4	11.6	1	0	2.2
OBD selection (%)						
U2 - shared alpha	18.2	28	1.2	0	0	52.6
U2 - dose-specific alpha	18.5	28	1.2	0	0	52.3
U2 - cycle-1 oracle	16	34.3	5.5	0.3	0	43.9
U3 - shared alpha	19	27.7	1	0	0	52.3
U3 - dose-specific alpha	18.8	27.1	1.3	0	0	52.8
U3 - cycle-1 oracle	15.8	34.7	5.2	0.2	0	44.1
Mean administrations						
U2 - shared alpha	10.2	7.4	3	0.4	0	-
U2 - dose-specific alpha	10.3	7.5	2.9	0.5	0	-
U2 - cycle-1 oracle	8.8	8.5	4.1	0.8	0.1	-
U3 - shared alpha	10.2	7.5	2.9	0.4	0	-
U3 - dose-specific alpha	10.2	7.5	2.9	0.5	0	-
U3 - cycle-1 oracle	8.8	8.5	4.1	0.8	0.1	-
Mean IPDE administrations						
U2 - shared alpha	2.9	3.2	0.9	0.2	0	-
U2 - dose-specific alpha	2.9	3.2	0.9	0.2	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.9	3.2	0.9	0.2	0	-
U3 - dose-specific alpha	2.9	3.2	0.9	0.2	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						760.5
U2 - dose-specific alpha						761.6
U2 - cycle-1 oracle						1235.1
U3 - shared alpha						762.4
U3 - dose-specific alpha						762.2
U3 - cycle-1 oracle						1234.6

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
MTD selection (%)						
U2 - shared alpha	53	36.6	7.2	0	0	3.2
U2 - dose-specific alpha	53	37	6.7	0.3	0	3
U2 - cycle-1 oracle	44.1	40.3	12.2	0.9	0	2.5
U3 - shared alpha	53.4	36.3	7	0.2	0	3.1
U3 - dose-specific alpha	53.2	36	7.4	0.2	0	3.2
U3 - cycle-1 oracle	44.2	40.2	12.5	0.6	0	2.5
OBD selection (%)						
U2 - shared alpha	74.2	20.8	1.7	0	0	3.3
U2 - dose-specific alpha	75.3	20	1.4	0	0	3.3
U2 - cycle-1 oracle	69.3	23.9	3.8	0.2	0	2.8
U3 - shared alpha	76.7	18.8	1.1	0	0	3.4
U3 - dose-specific alpha	76.6	18.5	1.4	0	0	3.5
U3 - cycle-1 oracle	70.9	23.2	3	0.2	0	2.7
Mean administrations						
U2 - shared alpha	19.5	7	2.6	0.3	0	-
U2 - dose-specific alpha	19.5	6.9	2.6	0.3	0	-
U2 - cycle-1 oracle	18	7.4	3.5	0.5	0	-
U3 - shared alpha	19.4	7	2.7	0.3	0	-
U3 - dose-specific alpha	19.5	6.9	2.6	0.3	0	-
U3 - cycle-1 oracle	18	7.4	3.5	0.5	0	-
Mean IPDE administrations						
U2 - shared alpha	5.1	2.3	0.7	0.1	0	-
U2 - dose-specific alpha	5.1	2.3	0.7	0.1	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	5.1	2.3	0.8	0.1	0	-
U3 - dose-specific alpha	5.1	2.3	0.7	0.1	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1165.7
U2 - dose-specific alpha						1164.8
U2 - cycle-1 oracle						1634.8
U3 - shared alpha						1164.9
U3 - dose-specific alpha						1164.9
U3 - cycle-1 oracle						1635.1
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
MTD selection (%)						
U2 - shared alpha	3.1	2.7	30.9	48.4	14.9	0
U2 - dose-specific alpha	3.1	3.1	30.6	48.4	14.8	0
U2 - cycle-1 oracle	2.1	2.3	24.2	46.4	25	0
U3 - shared alpha	3.1	2.5	31	48.3	15.1	0
U3 - dose-specific alpha	3.1	2.7	31.9	47.4	14.9	0
U3 - cycle-1 oracle	2.1	2.2	24	45.6	26.1	0
OBD selection (%)						
U2 - shared alpha	17.9	18.1	18.6	36.6	6.9	1.9
U2 - dose-specific alpha	19.2	17.4	18.6	36	6.9	1.9
U2 - cycle-1 oracle	20.6	14.9	14.5	36.7	11.6	1.7
U3 - shared alpha	23.9	20.4	14.1	33.9	5.7	2
U3 - dose-specific alpha	24.7	20.5	15.3	31.7	5.9	1.9
U3 - cycle-1 oracle	27.7	16.2	12.9	32	9.5	1.7
Mean administrations						
U2 - shared alpha	5.3	4.9	6.4	8.2	5	-
U2 - dose-specific alpha	5.2	4.9	6.5	8	5	-
U2 - cycle-1 oracle	4.9	4.7	5.9	8.3	5.9	-
U3 - shared alpha	5.3	5	6.3	8.1	5	-
U3 - dose-specific alpha	5.3	5	6.4	8	5	-
U3 - cycle-1 oracle	4.9	4.6	5.9	8.3	5.9	-
Mean IPDE administrations						
U2 - shared alpha	1	2.9	2.1	2.8	1.3	-
U2 - dose-specific alpha	1	2.9	2.1	2.8	1.3	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1	2.9	2	2.8	1.3	-
U3 - dose-specific alpha	1	2.9	2.1	2.8	1.3	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1081.9
U2 - dose-specific alpha						1081.9
U2 - cycle-1 oracle						1646
U3 - shared alpha						1080.9
U3 - dose-specific alpha						1080.4
U3 - cycle-1 oracle						1646

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
MTD selection (%)						
U2 - shared alpha	1.9	1.5	11.4	46.8	38.4	0
U2 - dose-specific alpha	1.9	1.5	11.4	46.9	38.3	0
U2 - cycle-1 oracle	1.5	1.3	8.5	40.5	48.2	0
U3 - shared alpha	1.9	1.5	12.2	46.7	37.7	0
U3 - dose-specific alpha	1.9	1.5	12	46.3	38.3	0
U3 - cycle-1 oracle	1.5	1.6	9	40.7	47.2	0
OBD selection (%)						
U2 - shared alpha	5.5	36.2	30.4	22.2	5.5	0.2
U2 - dose-specific alpha	5.1	37.3	28.8	23.2	5.4	0.2
U2 - cycle-1 oracle	7.6	36.5	26	22.9	6.8	0.2
U3 - shared alpha	7.5	41.6	28.5	18.2	4	0.2
U3 - dose-specific alpha	7.8	44.3	25.5	18.2	4	0.2
U3 - cycle-1 oracle	9.6	41.3	25.3	20.1	3.5	0.2
Mean administrations						
U2 - shared alpha	4.2	5.8	6.6	8	5.5	-
U2 - dose-specific alpha	4.1	5.9	6.5	8	5.4	-
U2 - cycle-1 oracle	4.1	6.1	6.4	7.7	5.8	-
U3 - shared alpha	4.1	5.8	6.6	8	5.5	-
U3 - dose-specific alpha	4.2	5.9	6.5	7.9	5.5	-
U3 - cycle-1 oracle	4.1	6	6.3	7.8	5.8	-
Mean IPDE administrations						
U2 - shared alpha	0.5	2.9	1.7	2.1	1.4	-
U2 - dose-specific alpha	0.5	2.9	1.7	2	1.4	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.5	2.9	1.7	2	1.4	-
U3 - dose-specific alpha	0.5	2.9	1.7	2	1.4	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1182.8
U2 - dose-specific alpha						1183.5
U2 - cycle-1 oracle						1662.3
U3 - shared alpha						1183.7
U3 - dose-specific alpha						1182.9
U3 - cycle-1 oracle						1662.3
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
MTD selection (%)						
U2 - shared alpha	0.7	0.8	10.2	43.7	44.6	0
U2 - dose-specific alpha	0.7	1	10.4	43.6	44.3	0
U2 - cycle-1 oracle	0.7	0.4	8.1	33.3	57.5	0
U3 - shared alpha	0.7	0.8	10.4	43.6	44.5	0
U3 - dose-specific alpha	0.7	0.9	10.8	43	44.6	0
U3 - cycle-1 oracle	0.7	0.4	8	34	56.9	0
OBD selection (%)						
U2 - shared alpha	9.5	14.2	35.9	29.5	10.2	0.7
U2 - dose-specific alpha	9.2	15.3	35.1	30	9.7	0.7
U2 - cycle-1 oracle	9.4	14.9	34.6	28	12.2	0.9
U3 - shared alpha	13.9	19.7	34.1	26	5.6	0.7
U3 - dose-specific alpha	13.3	20	34.1	25.9	6	0.7
U3 - cycle-1 oracle	14.7	17.7	33.6	24.4	8.8	0.8
Mean administrations						
U2 - shared alpha	3.9	4.4	8	7.5	6.1	-
U2 - dose-specific alpha	3.9	4.4	7.8	7.7	6.1	-
U2 - cycle-1 oracle	3.9	4.5	7.7	7.4	6.4	-
U3 - shared alpha	3.9	4.4	8	7.5	6.1	-
U3 - dose-specific alpha	3.9	4.4	7.8	7.7	6.1	-
U3 - cycle-1 oracle	3.9	4.5	7.7	7.4	6.4	-
Mean IPDE administrations						
U2 - shared alpha	0.4	2.9	2.5	2.8	2.1	-
U2 - dose-specific alpha	0.4	2.9	2.5	2.9	2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.4	2.9	2.5	2.8	2.1	-
U3 - dose-specific alpha	0.4	2.9	2.5	2.9	2.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1055.6
U2 - dose-specific alpha						1056.9
U2 - cycle-1 oracle						1656.6
U3 - shared alpha						1056.5
U3 - dose-specific alpha						1056.8
U3 - cycle-1 oracle						1656.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
MTD selection (%)						
U2 - shared alpha	0	0.5	12.5	48.5	38.5	0
U2 - dose-specific alpha	0	0.3	13.3	48.2	38.2	0
U2 - cycle-1 oracle	0	0.7	8.6	40.8	49.9	0
U3 - shared alpha	0	0.8	12.4	49.2	37.6	0
U3 - dose-specific alpha	0	0.6	13.3	47.7	38.4	0
U3 - cycle-1 oracle	0	0.8	9	40.2	50	0
OBD selection (%)						
U2 - shared alpha	7.9	8.8	15.7	42	25.5	0.1
U2 - dose-specific alpha	9	8.2	15	42.9	24.9	0
U2 - cycle-1 oracle	9	4.4	13.4	41.7	31.3	0.2
U3 - shared alpha	12	9.7	14.7	41.6	21.9	0.1
U3 - dose-specific alpha	13.2	10	15.1	39.3	22.3	0.1
U3 - cycle-1 oracle	14	6.2	15	41	23.6	0.2
Mean administrations						
U2 - shared alpha	3.5	3.6	5.3	8.1	9.4	-
U2 - dose-specific alpha	3.5	3.6	5.4	8.2	9.3	-
U2 - cycle-1 oracle	3.5	3.6	5.4	8.6	8.9	-
U3 - shared alpha	3.5	3.6	5.4	8.1	9.3	-
U3 - dose-specific alpha	3.5	3.7	5.4	8.1	9.3	-
U3 - cycle-1 oracle	3.5	3.6	5.4	8.6	8.9	-
Mean IPDE administrations						
U2 - shared alpha	0.2	2.6	1.7	3.1	2.3	-
U2 - dose-specific alpha	0.2	2.6	1.7	3	2.3	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.2	2.6	1.7	3	2.3	-
U3 - dose-specific alpha	0.2	2.6	1.7	3	2.3	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1104.4
U2 - dose-specific alpha						1105.2
U2 - cycle-1 oracle						1663.4
U3 - shared alpha						1104.3
U3 - dose-specific alpha						1104.3
U3 - cycle-1 oracle						1663.4

Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
MTD selection (%)						
U2 - shared alpha	4.4	16.5	49.4	21.9	7.8	0
U2 - dose-specific alpha	4.2	15.7	49.7	23.4	7	0
U2 - cycle-1 oracle	2.4	10.6	43.6	28.1	15.3	0
U3 - shared alpha	4.7	16	49.1	22.9	7.3	0
U3 - dose-specific alpha	4.4	15.4	50	23.1	7.1	0
U3 - cycle-1 oracle	2.5	10.1	43.2	29.3	14.9	0
OBD selection (%)						
U2 - shared alpha	14.9	15	47.5	13.8	3	5.8
U2 - dose-specific alpha	14.6	15.1	47.6	14.2	2.8	5.7
U2 - cycle-1 oracle	12.7	12.3	49.2	16.9	4.9	4
U3 - shared alpha	19.8	17.3	43.5	11.5	2.1	5.8
U3 - dose-specific alpha	19.4	16.8	44.3	11.7	1.9	5.9
U3 - cycle-1 oracle	19.6	13.1	43.6	16	3.5	4.2
Mean administrations						
U2 - shared alpha	5.3	6.3	10.3	4.9	2.3	-
U2 - dose-specific alpha	5.5	6.2	10.2	4.9	2.3	-
U2 - cycle-1 oracle	5	5.7	10.1	5.5	2.9	-
U3 - shared alpha	5.4	6.3	10.1	4.9	2.4	-
U3 - dose-specific alpha	5.4	6.2	10.2	4.8	2.3	-
U3 - cycle-1 oracle	5	5.7	10	5.6	3	-
Mean IPDE administrations						
U2 - shared alpha	1	3.5	3.3	1.7	0.7	-
U2 - dose-specific alpha	1.1	3.4	3.2	1.7	0.7	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.1	3.5	3.2	1.7	0.8	-
U3 - dose-specific alpha	1.1	3.4	3.3	1.7	0.8	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1032.8
U2 - dose-specific alpha						1029.4
U2 - cycle-1 oracle						1619.4
U3 - shared alpha						1030
U3 - dose-specific alpha						1029.6
U3 - cycle-1 oracle						1620.1

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
MTD selection (%)						
U2 - shared alpha	1.6	10	50.5	30.2	7.7	0
U2 - dose-specific alpha	1.6	10.7	50	28.9	8.8	0
U2 - cycle-1 oracle	1.3	6.3	45	31	16.4	0
U3 - shared alpha	1.6	10.5	50.6	29.7	7.6	0
U3 - dose-specific alpha	1.6	10.4	50.4	29.5	8.1	0
U3 - cycle-1 oracle	1.3	6.6	44.1	30.8	17.2	0
OBD selection (%)						
U2 - shared alpha	11.3	45.6	31.2	9.9	1.8	0.2
U2 - dose-specific alpha	11.7	44.6	32.1	9.8	1.6	0.2
U2 - cycle-1 oracle	12.6	41.1	30.3	12.6	3.2	0.2
U3 - shared alpha	14.2	49.3	27.2	8.1	1	0.2
U3 - dose-specific alpha	14.8	50.2	26	8	0.8	0.2
U3 - cycle-1 oracle	16	45.6	24.9	11.4	1.9	0.2
Mean administrations						
U2 - shared alpha	4.5	9.1	9.2	4.9	2.3	-
U2 - dose-specific alpha	4.6	9.1	9.3	4.8	2.2	-
U2 - cycle-1 oracle	4.5	8.7	8.9	5.1	2.7	-
U3 - shared alpha	4.5	9	9.3	4.9	2.2	-
U3 - dose-specific alpha	4.5	9.1	9.3	4.8	2.3	-
U3 - cycle-1 oracle	4.5	8.7	8.9	5.2	2.7	-
Mean IPDE administrations						
U2 - shared alpha	0.6	3.9	2.3	1.4	0.6	-
U2 - dose-specific alpha	0.7	3.9	2.3	1.4	0.7	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.7	3.8	2.3	1.5	0.6	-
U3 - dose-specific alpha	0.6	3.9	2.3	1.4	0.7	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1163.2
U2 - dose-specific alpha						1162.7
U2 - cycle-1 oracle						1662
U3 - shared alpha						1163.2
U3 - dose-specific alpha						1163.2
U3 - cycle-1 oracle						1662
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
MTD selection (%)						
U2 - shared alpha	36.8	37.4	17.8	6.1	1.2	0.7
U2 - dose-specific alpha	36.6	37.5	17.8	6.1	1.3	0.7
U2 - cycle-1 oracle	27.8	31.4	23.9	12	4.5	0.4
U3 - shared alpha	36.6	37.7	18.2	5.5	1.3	0.7
U3 - dose-specific alpha	37.2	36.6	18.8	5.4	1.4	0.6
U3 - cycle-1 oracle	27.9	31.7	24.4	10.7	4.9	0.4
OBD selection (%)						
U2 - shared alpha	18.6	37.3	10.2	3	0.8	30.1
U2 - dose-specific alpha	17.1	38.3	10.3	2.8	0.9	30.6
U2 - cycle-1 oracle	16.2	37.2	13.8	6	1.5	25.3
U3 - shared alpha	21.9	36.4	8.7	2.2	0.8	30
U3 - dose-specific alpha	22.9	36.5	8.2	2	0.8	29.6
U3 - cycle-1 oracle	19.8	37.6	10.5	5.3	1.1	25.7
Mean administrations						
U2 - shared alpha	9	8.4	5	1.9	0.7	-
U2 - dose-specific alpha	9	8.3	5	1.9	0.7	-
U2 - cycle-1 oracle	7.7	8.1	5.8	2.7	1.2	-
U3 - shared alpha	8.9	8.4	5	1.9	0.7	-
U3 - dose-specific alpha	8.9	8.5	4.9	1.8	0.7	-
U3 - cycle-1 oracle	7.7	8.2	5.8	2.8	1.2	-
Mean IPDE administrations						
U2 - shared alpha	2.4	4	1.6	0.8	0.2	-
U2 - dose-specific alpha	2.5	3.9	1.6	0.8	0.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.4	4	1.6	0.8	0.2	-
U3 - dose-specific alpha	2.4	4	1.6	0.7	0.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						874.2
U2 - dose-specific alpha						874.3
U2 - cycle-1 oracle						1424.4
U3 - shared alpha						875.5
U3 - dose-specific alpha						874.6
U3 - cycle-1 oracle						1423.9

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
MTD selection (%)						
U2 - shared alpha	29.3	46.5	20.5	3.2	0.3	0.2
U2 - dose-specific alpha	29.6	46.3	20.4	3.2	0.3	0.2
U2 - cycle-1 oracle	22.6	41	26.2	8	2.1	0.1
U3 - shared alpha	30.1	45.7	20.5	3.3	0.2	0.2
U3 - dose-specific alpha	29.5	46.7	20.2	3.1	0.3	0.2
U3 - cycle-1 oracle	22.1	41.3	26.9	7.8	1.8	0.1
OBD selection (%)						
U2 - shared alpha	23.4	45.4	10.4	0.9	0.1	19.8
U2 - dose-specific alpha	24	44.5	10.1	1.2	0.2	20
U2 - cycle-1 oracle	20.8	45.2	12.5	3.3	0.3	17.9
U3 - shared alpha	27.7	43.5	7.7	0.9	0	20.2
U3 - dose-specific alpha	28.1	42.5	8.2	0.9	0.1	20.2
U3 - cycle-1 oracle	27.1	42.8	9.4	2.2	0.2	18.3
Mean administrations						
U2 - shared alpha	8.3	10.8	5.5	1.5	0.5	-
U2 - dose-specific alpha	8.3	10.7	5.5	1.5	0.4	-
U2 - cycle-1 oracle	7.3	10.4	6.2	2.3	0.7	-
U3 - shared alpha	8.3	10.8	5.5	1.5	0.5	-
U3 - dose-specific alpha	8.3	10.8	5.6	1.4	0.5	-
U3 - cycle-1 oracle	7.4	10.2	6.3	2.3	0.8	-
Mean IPDE administrations						
U2 - shared alpha	2.1	5	1.7	0.6	0.2	-
U2 - dose-specific alpha	2.1	5	1.7	0.7	0.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.1	5	1.7	0.6	0.2	-
U3 - dose-specific alpha	2.1	5	1.8	0.6	0.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						926.5
U2 - cycle-1 oracle						924.9
U3 - shared alpha						1493.1
U3 - dose-specific alpha						926.9
U3 - cycle-1 oracle						927.1
						1490.2

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
MTD selection (%)						
U2 - shared alpha	70	11.8	3.2	0.2	0	14.8
U2 - dose-specific alpha	69.5	11.9	3.4	0.1	0	15.1
U2 - cycle-1 oracle	59.1	19.8	8	3.9	0.5	8.7
U3 - shared alpha	69.8	11.8	3.4	0	0	15
U3 - dose-specific alpha	69	12.3	3.1	0.1	0	15.5
U3 - cycle-1 oracle	58.9	20.3	7.3	3.9	0.5	9.1
OBD selection (%)						
U2 - shared alpha	14.6	4	0.9	0	0	80.5
U2 - dose-specific alpha	14.8	4.2	0.9	0	0	80.1
U2 - cycle-1 oracle	16.3	9.7	1.9	0.9	0.1	71.1
U3 - shared alpha	14.6	3.8	0.6	0	0	81
U3 - dose-specific alpha	14.4	4.5	0.7	0	0	80.4
U3 - cycle-1 oracle	15.7	10.5	1.7	0.4	0	71.7
Mean administrations						
U2 - shared alpha	11.4	4.8	1.4	0.6	0.1	-
U2 - dose-specific alpha	11.5	4.8	1.4	0.6	0.1	-
U2 - cycle-1 oracle	10.6	5.6	2.4	1.4	0.4	-
U3 - shared alpha	11.6	4.8	1.4	0.6	0.1	-
U3 - dose-specific alpha	11.6	4.8	1.4	0.6	0.1	-
U3 - cycle-1 oracle	10.8	5.5	2.3	1.3	0.4	-
Mean IPDE administrations						
U2 - shared alpha	3.1	2.6	0.5	0.3	0	-
U2 - dose-specific alpha	3.1	2.6	0.5	0.3	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.2	2.6	0.5	0.3	0	-
U3 - dose-specific alpha	3.2	2.6	0.5	0.3	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						646.3
U2 - cycle-1 oracle						646.9
U3 - shared alpha						1122.1
U3 - dose-specific alpha						647.5
U3 - cycle-1 oracle						645.1
						1118.6
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
MTD selection (%)						
U2 - shared alpha	10.1	36.2	42.4	10	1	0.3
U2 - dose-specific alpha	10.3	36.6	42	9.9	0.9	0.3
U2 - cycle-1 oracle	5.7	29.5	40.2	19.3	5	0.3
U3 - shared alpha	10	39.4	41.6	8.1	0.6	0.3
U3 - dose-specific alpha	9.9	38.9	42.1	8.1	0.7	0.3
U3 - cycle-1 oracle	5.5	31.2	40.7	18	4.3	0.3
OBD selection (%)						
U2 - shared alpha	27.5	24.9	37.1	7.4	0.4	2.7
U2 - dose-specific alpha	28.9	23.8	36.8	7.2	0.5	2.8
U2 - cycle-1 oracle	25.4	18.2	38.3	13	2.2	2.9
U3 - shared alpha	31.8	26.1	33.9	5.2	0.3	2.7
U3 - dose-specific alpha	33.7	24	34	5.3	0.3	2.7
U3 - cycle-1 oracle	29.1	19.4	36.2	11.1	1.5	2.7
Mean administrations						
U2 - shared alpha	10.5	8.3	8.3	2.3	0.2	-
U2 - dose-specific alpha	10.6	8.3	8.3	2.2	0.2	-
U2 - cycle-1 oracle	9.2	7	8.9	3.6	0.8	-
U3 - shared alpha	11	8.6	7.9	1.9	0.2	-
U3 - dose-specific alpha	11.1	8.5	7.9	2	0.2	-
U3 - cycle-1 oracle	9.6	7.2	8.7	3.4	0.7	-
Mean IPDE administrations						
U2 - shared alpha	2.7	3.7	2.4	0.7	0.1	-
U2 - dose-specific alpha	2.7	3.7	2.4	0.7	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.9	3.8	2.3	0.6	0.1	-
U3 - dose-specific alpha	2.9	3.8	2.3	0.6	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						1102.7
U2 - cycle-1 oracle						1102
U3 - shared alpha						1639.6
U3 - dose-specific alpha						1096.6
U3 - cycle-1 oracle						1096.7
						1641.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
MTD selection (%)						
U2 - shared alpha	14.2	38.7	34.9	11	0.9	0.3
U2 - dose-specific alpha	14.2	38.5	35.6	10.5	0.9	0.3
U2 - cycle-1 oracle	9.6	27.2	34.9	21.3	6.7	0.3
U3 - shared alpha	14	43.3	33.6	8.2	0.6	0.3
U3 - dose-specific alpha	14.1	43.1	34	7.7	0.8	0.3
U3 - cycle-1 oracle	9.4	31.3	34.3	18.9	5.8	0.3
OBD selection (%)						
U2 - shared alpha	9.5	50.2	23.3	6.1	0.5	10.4
U2 - dose-specific alpha	9.5	50.8	23.2	5.5	0.4	10.6
U2 - cycle-1 oracle	7.7	46.1	24.1	11.7	2.1	8.3
U3 - shared alpha	10	54.9	21	3.4	0.4	10.3
U3 - dose-specific alpha	9.9	55.2	20.6	3.3	0.4	10.6
U3 - cycle-1 oracle	9.6	51	20.6	9	1.6	8.2
Mean administrations						
U2 - shared alpha	7.5	12.4	6.1	2.3	0.3	-
U2 - dose-specific alpha	7.5	12.4	6.1	2.3	0.4	-
U2 - cycle-1 oracle	6.2	11.3	6.7	3.6	1	-
U3 - shared alpha	7.7	13.1	5.6	2	0.3	-
U3 - dose-specific alpha	7.7	13.1	5.6	2	0.3	-
U3 - cycle-1 oracle	6.3	12.2	6.3	3.2	0.8	-
Mean IPDE administrations						
U2 - shared alpha	1.5	5.1	1.8	0.8	0.1	-
U2 - dose-specific alpha	1.5	5.1	1.8	0.8	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.6	5.3	1.6	0.7	0.1	-
U3 - dose-specific alpha	1.6	5.3	1.7	0.7	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1057
U2 - dose-specific alpha						1055.9
U2 - cycle-1 oracle						1599.2
U3 - shared alpha						1053.7
U3 - dose-specific alpha						1052.7
U3 - cycle-1 oracle						1598.6
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
MTD selection (%)						
U2 - shared alpha	1	3.5	24.5	59	12	0
U2 - dose-specific alpha	1.1	2.9	24.8	59.2	12	0
U2 - cycle-1 oracle	0.7	2.5	16.3	54.7	25.8	0
U3 - shared alpha	1.1	3.4	24.1	62	9.4	0
U3 - dose-specific alpha	1.2	2.9	24.4	62.4	9.1	0
U3 - cycle-1 oracle	0.7	2.8	16.3	57.6	22.6	0
OBD selection (%)						
U2 - shared alpha	5.8	7.8	22.9	53.4	8.3	1.8
U2 - dose-specific alpha	6.1	7.6	23.6	52.6	8.3	1.8
U2 - cycle-1 oracle	6.2	6.3	16.4	54.2	15.3	1.6
U3 - shared alpha	9.7	7.9	21.2	53.6	5.7	1.9
U3 - dose-specific alpha	9.4	8.6	21.2	53.4	5.5	1.9
U3 - cycle-1 oracle	9.5	8.1	16.3	52.6	12.1	1.4
Mean administrations						
U2 - shared alpha	4.3	4.9	7.7	10.6	2.3	-
U2 - dose-specific alpha	4.3	5	7.7	10.5	2.4	-
U2 - cycle-1 oracle	4.1	4.6	6.9	10.7	3.6	-
U3 - shared alpha	4.5	5.2	7.7	10.5	2	-
U3 - dose-specific alpha	4.5	5.1	7.7	10.5	2	-
U3 - cycle-1 oracle	4.3	4.7	6.9	10.7	3.3	-
Mean IPDE administrations						
U2 - shared alpha	0.6	3.1	2.3	3.4	0.7	-
U2 - dose-specific alpha	0.5	3.2	2.3	3.4	0.7	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.6	3.3	2.3	3.4	0.6	-
U3 - dose-specific alpha	0.6	3.2	2.3	3.4	0.6	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1088.1
U2 - dose-specific alpha						1089
U2 - cycle-1 oracle						1653.4
U3 - shared alpha						1083.5
U3 - dose-specific alpha						1083.8
U3 - cycle-1 oracle						1653.8

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
MTD selection (%)						
U2 - shared alpha	1.9	6.7	19.6	38.5	33.3	0
U2 - dose-specific alpha	1.9	6.8	19.7	37.8	33.8	0
U2 - cycle-1 oracle	1.5	3.5	15.1	27.5	52.4	0
U3 - shared alpha	1.9	8.2	22.7	37.7	29.5	0
U3 - dose-specific alpha	1.9	8.2	23.1	37.2	29.6	0
U3 - cycle-1 oracle	1.5	4.8	18	30.3	45.4	0
OBD selection (%)						
U2 - shared alpha	4.9	34.6	26.1	23	9.5	1.9
U2 - dose-specific alpha	5.3	35	25.9	23.1	8.8	1.9
U2 - cycle-1 oracle	5.4	39	25.2	19.5	9.4	1.5
U3 - shared alpha	6.9	42.7	24.4	18.6	5.5	1.9
U3 - dose-specific alpha	7.6	41.6	24.7	18.2	6	1.9
U3 - cycle-1 oracle	8.5	44.4	23.5	16.3	5.8	1.5
Mean administrations						
U2 - shared alpha	4	8.1	7.4	6.9	3.2	-
U2 - dose-specific alpha	4.1	8.1	7.3	6.9	3.2	-
U2 - cycle-1 oracle	3.8	7.4	7.1	7.1	4.4	-
U3 - shared alpha	4.2	8.8	7.5	6.5	2.7	-
U3 - dose-specific alpha	4.2	8.8	7.5	6.5	2.7	-
U3 - cycle-1 oracle	3.9	8.2	7.2	6.8	3.8	-
Mean IPDE administrations						
U2 - shared alpha	0.4	4.1	2.4	2.6	1.2	-
U2 - dose-specific alpha	0.4	4.1	2.4	2.6	1.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.4	4.4	2.5	2.5	1	-
U3 - dose-specific alpha	0.4	4.4	2.5	2.5	1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1043.7
U2 - dose-specific alpha						1044.1
U2 - cycle-1 oracle						1650
U3 - shared alpha						1041.9
U3 - dose-specific alpha						1041.3
U3 - cycle-1 oracle						1650
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
MTD selection (%)						
U2 - shared alpha	58.5	18.8	4.3	0.7	0.1	17.6
U2 - dose-specific alpha	58.7	18.3	4.6	0.8	0.1	17.5
U2 - cycle-1 oracle	45.2	27.9	10.4	4.1	0.5	11.9
U3 - shared alpha	59.7	18.9	3.8	0.7	0.1	16.8
U3 - dose-specific alpha	59.6	18.2	4.4	0.7	0.1	17
U3 - cycle-1 oracle	45.4	28.9	9.2	4.1	0.5	11.9
OBD selection (%)						
U2 - shared alpha	63.6	13.1	2	0.5	0.1	20.7
U2 - dose-specific alpha	64.6	12.2	2.4	0.5	0.1	20.2
U2 - cycle-1 oracle	60.4	17	4.5	2.1	0.3	15.7
U3 - shared alpha	66.3	11.6	1.2	0.5	0.1	20.3
U3 - dose-specific alpha	65.5	11.9	1.4	0.4	0.1	20.7
U3 - cycle-1 oracle	62.9	16.2	3.8	1.6	0.2	15.3
Mean administrations						
U2 - shared alpha	18.9	5.7	1.4	0.5	0.1	-
U2 - dose-specific alpha	19	5.7	1.5	0.5	0.1	-
U2 - cycle-1 oracle	17.2	6	2.7	1.3	0.3	-
U3 - shared alpha	19.5	5.5	1.2	0.4	0.1	-
U3 - dose-specific alpha	19.4	5.5	1.2	0.4	0.1	-
U3 - cycle-1 oracle	17.7	5.9	2.5	1.1	0.2	-
Mean IPDE administrations						
U2 - shared alpha	4.9	2.4	0.5	0.2	0	-
U2 - dose-specific alpha	5	2.4	0.5	0.2	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	5.1	2.3	0.4	0.2	0	-
U3 - dose-specific alpha	5.1	2.3	0.4	0.2	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1031.9
U2 - dose-specific alpha						1033.8
U2 - cycle-1 oracle						1521.8
U3 - shared alpha						1033.9
U3 - dose-specific alpha						1032.7
U3 - cycle-1 oracle						1522.2

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
MTD selection (%)						
U2 - shared alpha	12.1	54.9	24.7	6.8	0.9	0.6
U2 - dose-specific alpha	12	55.7	24.5	6.4	0.9	0.5
U2 - cycle-1 oracle	6.8	41.6	29.2	15.4	6.6	0.4
U3 - shared alpha	12.2	61.3	20.3	5.1	0.6	0.5
U3 - dose-specific alpha	11.9	61.7	19.8	5.3	0.8	0.5
U3 - cycle-1 oracle	6.9	48.3	25.6	14.1	4.7	0.4
OBD selection (%)						
U2 - shared alpha	14.9	57.1	19.8	4.5	0.5	3.2
U2 - dose-specific alpha	15	57.6	20	3.9	0.5	3
U2 - cycle-1 oracle	11.8	48.3	23.5	10.3	3.3	2.8
U3 - shared alpha	16.6	60.9	15.4	3.4	0.3	3.4
U3 - dose-specific alpha	16.4	61	15.5	3.3	0.5	3.3
U3 - cycle-1 oracle	14.2	53.4	19.4	8.4	1.8	2.8
Mean administrations						
U2 - shared alpha	8.8	14.6	4.8	1.2	0.1	-
U2 - dose-specific alpha	8.9	14.7	4.7	1.1	0.1	-
U2 - cycle-1 oracle	7.6	12.7	5.8	2.7	0.8	-
U3 - shared alpha	9.2	15.3	4	0.9	0.1	-
U3 - dose-specific alpha	9.2	15.3	4	0.9	0.1	-
U3 - cycle-1 oracle	7.8	13.6	5.2	2.4	0.6	-
Mean IPDE administrations						
U2 - shared alpha	1.9	5.3	1.3	0.4	0	-
U2 - dose-specific alpha	1.9	5.3	1.3	0.4	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2	5.5	1.1	0.3	0	-
U3 - dose-specific alpha	2	5.5	1.1	0.3	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						1137.2
U2 - cycle-1 oracle						1137.9
U3 - shared alpha						1639.8
U3 - dose-specific alpha						1134
U3 - cycle-1 oracle						1132.6
						1639.8
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
MTD selection (%)						
U2 - shared alpha	66.3	8.6	1.6	0.3	0	23.2
U2 - dose-specific alpha	66.1	8.7	1.8	0.2	0	23.2
U2 - cycle-1 oracle	58.9	17.8	7	1.8	0.5	14
U3 - shared alpha	66.7	8.5	1.7	0.3	0	22.8
U3 - dose-specific alpha	66.2	8.8	1.6	0.3	0	23.1
U3 - cycle-1 oracle	58.5	18.4	6.7	1.5	0.6	14.3
OBD selection (%)						
U2 - shared alpha	29.6	5.5	0.4	0.2	0	64.3
U2 - dose-specific alpha	29.9	5	0.6	0.1	0	64.4
U2 - cycle-1 oracle	27.6	11.9	3.1	0.9	0.2	56.3
U3 - shared alpha	30.2	5	0.5	0.1	0	64.2
U3 - dose-specific alpha	30.5	5.2	0.2	0.1	0	64
U3 - cycle-1 oracle	28	12.1	2.5	0.6	0.3	56.5
Mean administrations						
U2 - shared alpha	14.1	4.9	1.1	0.4	0.1	-
U2 - dose-specific alpha	14.1	4.9	1.2	0.4	0.1	-
U2 - cycle-1 oracle	12.6	5.8	2.3	1.1	0.2	-
U3 - shared alpha	14.3	4.8	1.1	0.4	0.1	-
U3 - dose-specific alpha	14.3	4.8	1.1	0.4	0.1	-
U3 - cycle-1 oracle	12.9	5.7	2.2	1.1	0.2	-
Mean IPDE administrations						
U2 - shared alpha	3.9	2.5	0.4	0.2	0	-
U2 - dose-specific alpha	3.9	2.4	0.4	0.2	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	4	2.4	0.4	0.2	0	-
U3 - dose-specific alpha	4	2.4	0.4	0.2	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						746.1
U2 - cycle-1 oracle						745.9
U3 - shared alpha						1218.7
U3 - dose-specific alpha						746.4
U3 - cycle-1 oracle						747.1
						1219.6

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
MTD selection (%)						
U2 - shared alpha	14.6	18.4	20.6	26.4	19.7	0.3
U2 - dose-specific alpha	15	17.4	20.8	26.6	19.9	0.3
U2 - cycle-1 oracle	9.3	10.9	17.5	25.4	36.6	0.3
U3 - shared alpha	14.9	18.8	20.9	27.2	17.9	0.3
U3 - dose-specific alpha	14.7	18.7	21.2	27.3	17.8	0.3
U3 - cycle-1 oracle	9.3	11.8	16.8	26.3	35.5	0.3
OBD selection (%)						
U2 - shared alpha	15	13.9	22.4	15.3	14.7	18.7
U2 - dose-specific alpha	14.7	13.6	22.1	15.2	15.4	19
U2 - cycle-1 oracle	11.5	10.6	21.6	16.1	26.1	14.1
U3 - shared alpha	18.3	15.8	20.7	13.7	12.2	19.3
U3 - dose-specific alpha	18.4	16.3	20.2	14.1	12.1	18.9
U3 - cycle-1 oracle	15.2	13.6	19.9	14.4	22.6	14.3
Mean administrations						
U2 - shared alpha	7.5	6.2	6	4.9	3.1	-
U2 - dose-specific alpha	7.4	6.2	6	4.9	3.1	-
U2 - cycle-1 oracle	6.2	5.7	6	5.5	4.8	-
U3 - shared alpha	7.7	6.3	5.9	4.9	2.8	-
U3 - dose-specific alpha	7.6	6.3	5.9	4.9	2.9	-
U3 - cycle-1 oracle	6.5	5.8	6	5.4	4.7	-
Mean IPDE administrations						
U2 - shared alpha	1.7	3.5	2	1.9	1	-
U2 - dose-specific alpha	1.7	3.6	2	1.9	1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.8	3.6	2	1.9	0.9	-
U3 - dose-specific alpha	1.8	3.6	2	1.9	0.9	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						954.1
U2 - dose-specific alpha						954.9
U2 - cycle-1 oracle						1569.8
U3 - shared alpha						951.5
U3 - dose-specific alpha						952.9
U3 - cycle-1 oracle						1568.9
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
MTD selection (%)						
U2 - shared alpha	50.8	25.7	15.1	4.1	0.1	4.2
U2 - dose-specific alpha	51.2	25.4	14.9	4.2	0.1	4.2
U2 - cycle-1 oracle	37.3	25.7	20.3	11.5	2.8	2.4
U3 - shared alpha	51.8	25.9	14.7	3.4	0.1	4.1
U3 - dose-specific alpha	51.3	26.7	14.5	3.4	0.2	3.9
U3 - cycle-1 oracle	37.7	27	20.5	10.1	2.4	2.3
OBD selection (%)						
U2 - shared alpha	10.6	28.5	12.3	2.3	0.1	46.2
U2 - dose-specific alpha	10.1	28.6	11.6	2.5	0.1	47.1
U2 - cycle-1 oracle	5.7	34.8	14.9	6.7	1.5	36.4
U3 - shared alpha	11	29.5	10.5	2	0.1	46.9
U3 - dose-specific alpha	10.3	29.8	10.6	2.2	0.2	46.9
U3 - cycle-1 oracle	6	37.7	13.6	5.2	0.9	36.6
Mean administrations						
U2 - shared alpha	9.6	9.2	3.4	1.1	0.2	-
U2 - dose-specific alpha	9.6	9.2	3.4	1.1	0.2	-
U2 - cycle-1 oracle	7.7	9.8	4.6	2.1	0.6	-
U3 - shared alpha	9.7	9.3	3.3	1	0.1	-
U3 - dose-specific alpha	9.6	9.4	3.3	1	0.1	-
U3 - cycle-1 oracle	7.8	10.2	4.4	2	0.5	-
Mean IPDE administrations						
U2 - shared alpha	2.2	4.1	1	0.4	0.1	-
U2 - dose-specific alpha	2.2	4.1	1	0.4	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.3	4.1	1	0.4	0	-
U3 - dose-specific alpha	2.2	4.1	1	0.4	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						865
U2 - dose-specific alpha						865.5
U2 - cycle-1 oracle						1376.5
U3 - shared alpha						864
U3 - dose-specific alpha						864.2
U3 - cycle-1 oracle						1375.7

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
MTD selection (%)						
U2 - shared alpha	52.3	36.8	6.4	0	0	4.5
U2 - dose-specific alpha	51.4	37.8	6.1	0	0	4.7
U2 - cycle-1 oracle	36	44.1	14.9	2.2	0	2.8
U3 - shared alpha	52.2	37.4	5.8	0	0	4.6
U3 - dose-specific alpha	52.4	37.5	5.8	0	0	4.3
U3 - cycle-1 oracle	34.7	45.2	15.2	2	0	2.9
OBD selection (%)						
U2 - shared alpha	65.5	17.4	1.3	0	0	15.8
U2 - dose-specific alpha	65.2	18	1.3	0	0	15.5
U2 - cycle-1 oracle	58	22.5	4.5	0.3	0	14.7
U3 - shared alpha	66.6	16.7	1	0	0	15.7
U3 - dose-specific alpha	68	15.7	1.2	0	0	15.1
U3 - cycle-1 oracle	60.8	19.9	3.8	0.3	0	15.2
Mean administrations						
U2 - shared alpha	17.7	7.2	2.2	0.7	0	-
U2 - dose-specific alpha	17.7	7.2	2.2	0.8	0.1	-
U2 - cycle-1 oracle	15.7	7.7	3.2	1.3	0.2	-
U3 - shared alpha	18	7.1	2.1	0.7	0	-
U3 - dose-specific alpha	18.1	7	2	0.7	0	-
U3 - cycle-1 oracle	16.2	7.4	3.1	1.3	0.2	-
Mean IPDE administrations						
U2 - shared alpha	5.2	3.4	0.7	0.3	0	-
U2 - dose-specific alpha	5.2	3.4	0.7	0.3	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	5.3	3.4	0.6	0.3	0	-
U3 - dose-specific alpha	5.4	3.3	0.7	0.3	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						1005.3
U2 - cycle-1 oracle						1006.7
U3 - shared alpha						1564.3
U3 - dose-specific alpha						1003.5
U3 - cycle-1 oracle						1004.3
						1561.3
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
MTD selection (%)						
U2 - shared alpha	1.3	2.6	26.3	54.9	14.9	0
U2 - dose-specific alpha	1.3	2.4	27.1	54	15.2	0
U2 - cycle-1 oracle	0.9	2.7	16.2	50	30.2	0
U3 - shared alpha	1.4	2.3	26.8	55.2	14.3	0
U3 - dose-specific alpha	1.2	2.7	25.9	56.4	13.8	0
U3 - cycle-1 oracle	0.9	2.6	15.5	52.8	28.2	0
OBD selection (%)						
U2 - shared alpha	11.5	14.1	13.4	46.4	12.2	2.4
U2 - dose-specific alpha	10.8	15.3	13.3	45.8	12.4	2.4
U2 - cycle-1 oracle	11.4	11.9	8.1	44.6	21.9	2.1
U3 - shared alpha	16.5	15.4	11.8	43	10.9	2.4
U3 - dose-specific alpha	16.6	15.7	10.2	44.3	10.9	2.3
U3 - cycle-1 oracle	14.9	14.1	7.9	41.7	19.1	2.3
Mean administrations						
U2 - shared alpha	4.7	5.8	6.3	9.4	3.6	-
U2 - dose-specific alpha	4.7	5.9	6.2	9.4	3.6	-
U2 - cycle-1 oracle	4.4	5.2	5.6	9.5	5.1	-
U3 - shared alpha	5	6	6	9.3	3.4	-
U3 - dose-specific alpha	5	6	6.1	9.2	3.4	-
U3 - cycle-1 oracle	4.6	5.2	5.5	9.6	4.8	-
Mean IPDE administrations						
U2 - shared alpha	0.7	3.6	1.9	3.3	1	-
U2 - dose-specific alpha	0.7	3.6	1.9	3.3	1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.9	3.6	1.8	3.3	1	-
U3 - dose-specific alpha	0.9	3.7	1.8	3.3	0.9	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						1060.5
U2 - cycle-1 oracle						1060.5
U3 - shared alpha						1651.5
U3 - dose-specific alpha						1057
U3 - cycle-1 oracle						1055.6
						1650.3

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
MTD selection (%)						
U2 - shared alpha	50.4	24.5	14.6	7	0.9	2.6
U2 - dose-specific alpha	50.3	24.8	14.6	6.7	0.9	2.7
U2 - cycle-1 oracle	34.7	24.5	19.8	15.9	3.3	1.8
U3 - shared alpha	50	25.1	15.7	6.1	0.5	2.6
U3 - dose-specific alpha	50.9	24.5	15.4	5.9	0.7	2.6
U3 - cycle-1 oracle	35.3	24.9	19.8	15.5	2.7	1.8
OBD selection (%)						
U2 - shared alpha	12	29	10	3.7	0.4	44.9
U2 - dose-specific alpha	12.4	29.4	10	3.3	0.5	44.4
U2 - cycle-1 oracle	7	36.4	14.6	6.4	1.6	34
U3 - shared alpha	13.2	29.8	9.9	2.4	0.4	44.3
U3 - dose-specific alpha	13.1	30.2	8.9	2.5	0.4	44.9
U3 - cycle-1 oracle	9.3	36.5	14.6	4.5	1.2	33.9
Mean administrations						
U2 - shared alpha	9.7	8.5	3.8	1.8	0.3	-
U2 - dose-specific alpha	9.7	8.5	3.7	1.7	0.3	-
U2 - cycle-1 oracle	7.8	9.1	4.8	2.8	0.9	-
U3 - shared alpha	9.8	8.8	3.6	1.6	0.3	-
U3 - dose-specific alpha	9.8	8.8	3.6	1.5	0.3	-
U3 - cycle-1 oracle	7.9	9.3	4.7	2.6	0.7	-
Mean IPDE administrations						
U2 - shared alpha	2.3	4	1.2	0.7	0.1	-
U2 - dose-specific alpha	2.3	4	1.1	0.7	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.4	4.2	1.1	0.6	0.1	-
U3 - dose-specific alpha	2.4	4.2	1.1	0.6	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						859.7
U2 - dose-specific alpha						857.5
U2 - cycle-1 oracle						1404.3
U3 - shared alpha						856.3
U3 - dose-specific alpha						855.9
U3 - cycle-1 oracle						1400.1
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
MTD selection (%)						
U2 - shared alpha	4.3	12	30.6	37.6	15.5	0
U2 - dose-specific alpha	4.3	12	30.5	37.5	15.7	0
U2 - cycle-1 oracle	2.5	8.3	22.5	36.3	30.4	0
U3 - shared alpha	4.3	12.3	30.6	37.7	15.1	0
U3 - dose-specific alpha	4.3	11.9	30.3	38.4	15.1	0
U3 - cycle-1 oracle	2.5	9	21.8	37.7	29	0
OBD selection (%)						
U2 - shared alpha	17.3	17.9	19.7	28.1	8.9	8.1
U2 - dose-specific alpha	17.5	19.6	18	28	8.6	8.3
U2 - cycle-1 oracle	17.2	15.5	14.1	32.1	14.9	6.2
U3 - shared alpha	22.7	21.7	15.2	25.7	6.5	8.2
U3 - dose-specific alpha	22.3	21.3	15	26.2	7	8.2
U3 - cycle-1 oracle	21.5	19	12.7	29.4	11.2	6.2
Mean administrations						
U2 - shared alpha	6.3	6.8	6.4	6.9	2.9	-
U2 - dose-specific alpha	6.2	6.8	6.4	6.9	2.9	-
U2 - cycle-1 oracle	5.7	6.1	5.8	7.7	4.1	-
U3 - shared alpha	6.6	7	6.2	6.7	2.7	-
U3 - dose-specific alpha	6.7	6.9	6.1	6.8	2.7	-
U3 - cycle-1 oracle	6	6.1	5.6	7.8	3.8	-
Mean IPDE administrations						
U2 - shared alpha	1.4	3.9	2.1	2.7	1	-
U2 - dose-specific alpha	1.4	3.9	2.1	2.7	1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.5	4	2	2.6	0.9	-
U3 - dose-specific alpha	1.6	4	1.9	2.6	0.9	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1001.5
U2 - dose-specific alpha						1000.2
U2 - cycle-1 oracle						1631.6
U3 - shared alpha						997.9
U3 - dose-specific alpha						998.5
U3 - cycle-1 oracle						1631.2

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
MTD selection (%)						
U2 - shared alpha	11.2	37.4	35.7	13.3	2.2	0.2
U2 - dose-specific alpha	12	35.7	36.1	14	2	0.2
U2 - cycle-1 oracle	8	25.8	34.6	22.9	8.5	0.2
U3 - shared alpha	11.6	36.6	36.3	13.4	1.9	0.2
U3 - dose-specific alpha	12	35.7	35.9	14.2	2	0.2
U3 - cycle-1 oracle	8.1	26.7	33.6	22.7	8.7	0.2
OBD selection (%)						
U2 - shared alpha	40.6	29.8	11.4	4.2	0.3	13.7
U2 - dose-specific alpha	41.9	29.3	10.9	3.5	0.2	14.2
U2 - cycle-1 oracle	40.6	28.5	11.8	6.3	0.3	12.5
U3 - shared alpha	44.8	30.7	7.4	2.8	0.3	14
U3 - dose-specific alpha	45.5	30.5	6.7	2.9	0.3	14.1
U3 - cycle-1 oracle	45.7	29.1	9.1	3.1	0.2	12.8
Mean administrations						
U2 - shared alpha	10.7	8.7	5.3	3	0.8	-
U2 - dose-specific alpha	10.6	8.7	5.4	3	0.8	-
U2 - cycle-1 oracle	9.4	8.1	5.8	4	1.5	-
U3 - shared alpha	11.1	8.6	5.1	2.9	0.8	-
U3 - dose-specific alpha	11.1	8.6	5	2.9	0.8	-
U3 - cycle-1 oracle	10	8	5.5	3.8	1.4	-
Mean IPDE administrations						
U2 - shared alpha	3.1	4.4	1.8	1.3	0.3	-
U2 - dose-specific alpha	3.1	4.5	1.8	1.3	0.3	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.3	4.4	1.7	1.2	0.3	-
U3 - dose-specific alpha	3.3	4.4	1.7	1.2	0.3	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						960.3
U2 - dose-specific alpha						961.2
U2 - cycle-1 oracle						1595.3
U3 - shared alpha						959.7
U3 - dose-specific alpha						959
U3 - cycle-1 oracle						1596.2
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
MTD selection (%)						
U2 - shared alpha	0	1.7	2.7	14.9	80.7	0
U2 - dose-specific alpha	0	1.3	3	14.4	81.3	0
U2 - cycle-1 oracle	0	1.6	1.1	10.2	87.1	0
U3 - shared alpha	0	2.5	4.3	16.5	76.7	0
U3 - dose-specific alpha	0	2.7	4.8	15.9	76.6	0
U3 - cycle-1 oracle	0	1.6	2.4	10.5	85.5	0
OBD selection (%)						
U2 - shared alpha	35.1	30.2	15.5	12.8	6.4	0
U2 - dose-specific alpha	36.1	29.3	14.1	13.2	7.3	0
U2 - cycle-1 oracle	39.6	25.4	15.5	12.7	6.8	0
U3 - shared alpha	39.6	32	13.8	10.4	4.2	0
U3 - dose-specific alpha	40.5	32.2	12.7	10.2	4.4	0
U3 - cycle-1 oracle	46.1	24.2	14.6	10.8	4.3	0
Mean administrations						
U2 - shared alpha	4.4	5.3	6.1	8.2	6	-
U2 - dose-specific alpha	4.3	5.3	6.1	8.3	6	-
U2 - cycle-1 oracle	4.5	4.9	6.1	8.4	6.1	-
U3 - shared alpha	4.7	5.8	6.2	7.9	5.4	-
U3 - dose-specific alpha	4.8	5.8	6.1	8	5.3	-
U3 - cycle-1 oracle	4.6	5.2	6.2	8.3	5.7	-
Mean IPDE administrations						
U2 - shared alpha	0.6	3	2.3	3.5	2.3	-
U2 - dose-specific alpha	0.5	3	2.3	3.5	2.3	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.7	3.2	2.3	3.4	2.1	-
U3 - dose-specific alpha	0.7	3.2	2.3	3.4	2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1007.8
U2 - dose-specific alpha						1007.9
U2 - cycle-1 oracle						1664.1
U3 - shared alpha						1005.5
U3 - dose-specific alpha						1005.5
U3 - cycle-1 oracle						1664.1

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
MTD selection (%)						
U2 - shared alpha	7.5	74.3	17.4	0.7	0	0.1
U2 - dose-specific alpha	6.9	75	17.4	0.6	0	0.1
U2 - cycle-1 oracle	5.5	60.2	30.2	3.6	0.4	0.1
U3 - shared alpha	7.5	78.3	13.5	0.6	0	0.1
U3 - dose-specific alpha	7.8	77.2	14.4	0.5	0	0.1
U3 - cycle-1 oracle	5.5	64.3	27.1	3	0	0.1
OBD selection (%)						
U2 - shared alpha	20.4	68.5	10.2	0.5	0	0.4
U2 - dose-specific alpha	21.8	68.2	9.1	0.4	0	0.5
U2 - cycle-1 oracle	19.7	61.1	16.8	1.7	0.1	0.6
U3 - shared alpha	23.6	68.7	7	0.3	0	0.4
U3 - dose-specific alpha	23.5	67.9	7.8	0.3	0	0.5
U3 - cycle-1 oracle	22.8	62.4	12.6	1.5	0	0.7
Mean administrations						
U2 - shared alpha	10.1	16.2	3.3	0.4	0	-
U2 - dose-specific alpha	10.1	16.2	3.2	0.3	0	-
U2 - cycle-1 oracle	9.1	14.3	5.1	1.2	0.2	-
U3 - shared alpha	10.8	16.4	2.5	0.3	0	-
U3 - dose-specific alpha	10.8	16.3	2.6	0.2	0	-
U3 - cycle-1 oracle	9.6	14.7	4.5	1	0.1	-
Mean IPDE administrations						
U2 - shared alpha	2.3	5.4	1	0.1	0	-
U2 - dose-specific alpha	2.3	5.4	0.9	0.1	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.5	5.5	0.7	0.1	0	-
U3 - dose-specific alpha	2.5	5.5	0.7	0.1	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1171
U2 - dose-specific alpha						1170
U2 - cycle-1 oracle						1658.1
U3 - shared alpha						1165.5
U3 - dose-specific alpha						1164.7
U3 - cycle-1 oracle						1657.6
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
MTD selection (%)						
U2 - shared alpha	0.1	0.1	3.7	31.2	64.9	0
U2 - dose-specific alpha	0.1	0	3.4	31.8	64.7	0
U2 - cycle-1 oracle	0	0.2	1.8	22.8	75.2	0
U3 - shared alpha	0.1	0.2	3.2	32.1	64.4	0
U3 - dose-specific alpha	0.1	0.1	3.1	31.9	64.8	0
U3 - cycle-1 oracle	0	0.2	1.8	22.6	75.4	0
OBD selection (%)						
U2 - shared alpha	9.2	10.8	11	19	49.9	0.1
U2 - dose-specific alpha	8.7	12.1	10.6	19.3	49.2	0.1
U2 - cycle-1 oracle	12	9.9	8.7	14.8	54.6	0
U3 - shared alpha	14.7	13.3	10.4	17.6	43.9	0.1
U3 - dose-specific alpha	15.4	12.6	11.3	15.9	44.7	0.1
U3 - cycle-1 oracle	17.5	14.3	9.4	11.7	47.1	0
Mean administrations						
U2 - shared alpha	3.4	4.1	5.3	7.4	9.8	-
U2 - dose-specific alpha	3.4	4.1	5.3	7.4	9.7	-
U2 - cycle-1 oracle	3.4	4.1	5.2	7	10.2	-
U3 - shared alpha	3.5	4.2	5.4	7.6	9.2	-
U3 - dose-specific alpha	3.5	4.2	5.5	7.6	9.2	-
U3 - cycle-1 oracle	3.5	4.2	5.3	7	9.9	-
Mean IPDE administrations						
U2 - shared alpha	0.2	2.8	1.8	3	2.9	-
U2 - dose-specific alpha	0.2	2.8	1.8	3	2.9	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.2	2.9	1.9	3.1	2.7	-
U3 - dose-specific alpha	0.3	2.9	1.9	3.1	2.7	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1057.1
U2 - dose-specific alpha						1057.3
U2 - cycle-1 oracle						1664.1
U3 - shared alpha						1052.2
U3 - dose-specific alpha						1052.2
U3 - cycle-1 oracle						1664.1

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
MTD selection (%)						
U2 - shared alpha	22	52.1	24.4	1.5	0	0
U2 - dose-specific alpha	22.3	51.8	24.3	1.6	0	0
U2 - cycle-1 oracle	17.2	43.9	33	5.9	0	0
U3 - shared alpha	21.9	52.5	24	1.6	0	0
U3 - dose-specific alpha	22	51.6	24.7	1.7	0	0
U3 - cycle-1 oracle	17.1	43.6	33.2	6.1	0	0
OBD selection (%)						
U2 - shared alpha	17	55	12.5	0.3	0	15.2
U2 - dose-specific alpha	16.3	54.2	13.4	0.4	0	15.7
U2 - cycle-1 oracle	11.9	54	17.6	2	0	14.5
U3 - shared alpha	18.2	54.5	10.9	0.3	0	16.1
U3 - dose-specific alpha	18.2	54.8	11.2	0.2	0	15.6
U3 - cycle-1 oracle	13.6	55.4	15.3	1.7	0	14
Mean administrations						
U2 - shared alpha	8.8	12.9	5.3	1.4	0.1	-
U2 - dose-specific alpha	8.7	12.8	5.3	1.4	0.1	-
U2 - cycle-1 oracle	7.4	12.2	6.3	2.2	0.4	-
U3 - shared alpha	9	13	5.1	1.3	0.1	-
U3 - dose-specific alpha	9	13.1	5	1.2	0.1	-
U3 - cycle-1 oracle	7.6	12.6	6	2	0.3	-
Mean IPDE administrations						
U2 - shared alpha	2	5.6	1.6	0.6	0	-
U2 - dose-specific alpha	2	5.6	1.6	0.6	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.1	5.7	1.5	0.5	0	-
U3 - dose-specific alpha	2.1	5.7	1.5	0.5	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						1017.3
U2 - cycle-1 oracle						1017.5
U3 - shared alpha						1579
U3 - dose-specific alpha						1012.5
U3 - cycle-1 oracle						1013.1
						1579.1
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
MTD selection (%)						
U2 - shared alpha	2.6	19.4	53.1	22.2	2.7	0
U2 - dose-specific alpha	2.6	20.8	52.2	21.9	2.5	0
U2 - cycle-1 oracle	2.2	12.4	43.9	32.8	8.7	0
U3 - shared alpha	2.4	19.9	53.3	22	2.4	0
U3 - dose-specific alpha	2.4	20.2	53.7	21.5	2.2	0
U3 - cycle-1 oracle	2.3	13	43.8	33.1	7.8	0
OBD selection (%)						
U2 - shared alpha	9.2	51.4	28.2	8.1	0.6	2.5
U2 - dose-specific alpha	8.7	52.4	28.8	7	0.7	2.4
U2 - cycle-1 oracle	9.1	48	26.9	11.8	1.7	2.5
U3 - shared alpha	11	56.4	23.1	6.4	0.5	2.6
U3 - dose-specific alpha	12.5	55.9	22.3	6.4	0.5	2.4
U3 - cycle-1 oracle	11.5	50.3	24.2	10.2	1.1	2.7
Mean administrations						
U2 - shared alpha	5.1	11.2	8.2	4.2	1	-
U2 - dose-specific alpha	5.1	11.1	8.2	4.2	1	-
U2 - cycle-1 oracle	4.7	10.2	8.2	5	1.7	-
U3 - shared alpha	5.4	11.7	7.8	3.8	0.9	-
U3 - dose-specific alpha	5.4	11.7	7.8	3.9	0.9	-
U3 - cycle-1 oracle	4.9	10.6	7.9	4.7	1.6	-
Mean IPDE administrations						
U2 - shared alpha	0.8	5.3	2.6	1.6	0.4	-
U2 - dose-specific alpha	0.8	5.2	2.6	1.6	0.4	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.9	5.5	2.5	1.5	0.3	-
U3 - dose-specific alpha	0.9	5.5	2.5	1.5	0.3	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						1043.4
U2 - cycle-1 oracle						1045
U3 - shared alpha						1645.9
U3 - dose-specific alpha						1039.6
U3 - cycle-1 oracle						1041
						1645.6

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
MTD selection (%)						
U2 - shared alpha	47.5	0.8	0	0	0	51.7
U2 - dose-specific alpha	48	1	0	0	0	51
U2 - cycle-1 oracle	57.3	3.8	0	0	0	38.9
U3 - shared alpha	47.8	0.8	0	0	0	51.4
U3 - dose-specific alpha	48	0.8	0	0	0	51.2
U3 - cycle-1 oracle	56.9	4.1	0	0	0	39
OBD selection (%)						
U2 - shared alpha	45.4	0.1	0	0	0	54.5
U2 - dose-specific alpha	45.4	0.2	0	0	0	54.4
U2 - cycle-1 oracle	57.4	1.5	0	0	0	41.1
U3 - shared alpha	45.5	0.1	0	0	0	54.4
U3 - dose-specific alpha	45.3	0.1	0	0	0	54.6
U3 - cycle-1 oracle	57.5	1.6	0	0	0	40.9
Mean administrations						
U2 - shared alpha	19.2	1.9	0	0	0	-
U2 - dose-specific alpha	19.2	1.9	0	0	0	-
U2 - cycle-1 oracle	20.2	2.5	0.2	0	0	-
U3 - shared alpha	19.3	1.8	0	0	0	-
U3 - dose-specific alpha	19.2	1.9	0	0	0	-
U3 - cycle-1 oracle	20.3	2.4	0.2	0	0	-
Mean IPDE administrations						
U2 - shared alpha	3.7	0.8	0	0	0	-
U2 - dose-specific alpha	3.7	0.8	0	0	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	3.7	0.8	0	0	0	-
U3 - dose-specific alpha	3.7	0.8	0	0	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						914.6
U2 - cycle-1 oracle						916.4
U3 - shared alpha						1274.5
U3 - dose-specific alpha						914.7
U3 - cycle-1 oracle						915.1
						1272.7
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
MTD selection (%)						
U2 - shared alpha	18.6	59.3	21	1.1	0	0
U2 - dose-specific alpha	18.2	60.8	19.9	1.1	0	0
U2 - cycle-1 oracle	15	50.6	30.7	3.7	0	0
U3 - shared alpha	17.7	59.8	21.1	1.4	0	0
U3 - dose-specific alpha	18.1	59.1	21.4	1.4	0	0
U3 - cycle-1 oracle	14.7	49.2	32	4.1	0	0
OBD selection (%)						
U2 - shared alpha	17.5	60.4	8	0.9	0	13.2
U2 - dose-specific alpha	18.3	61.1	7	0.9	0	12.7
U2 - cycle-1 oracle	14.4	57.9	11.3	2.5	0	13.9
U3 - shared alpha	18.4	61.3	6.4	1.2	0	12.7
U3 - dose-specific alpha	19.4	60.5	6.5	1.1	0	12.5
U3 - cycle-1 oracle	16.6	57.5	9.7	2.6	0	13.6
Mean administrations						
U2 - shared alpha	8.9	13.6	4.9	1.5	0.1	-
U2 - dose-specific alpha	8.9	13.7	4.8	1.4	0.1	-
U2 - cycle-1 oracle	8.1	12.9	5.5	2.1	0.2	-
U3 - shared alpha	9.2	13.7	4.6	1.4	0	-
U3 - dose-specific alpha	9.2	13.6	4.7	1.4	0	-
U3 - cycle-1 oracle	8.5	12.8	5.3	2	0.2	-
Mean IPDE administrations						
U2 - shared alpha	2.1	6	1.5	0.6	0	-
U2 - dose-specific alpha	2.1	6.1	1.4	0.6	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.3	6.1	1.4	0.5	0	-
U3 - dose-specific alpha	2.3	6.1	1.4	0.5	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						1025.1
U2 - cycle-1 oracle						1023.5
U3 - shared alpha						1598
U3 - dose-specific alpha						1021.1
U3 - cycle-1 oracle						1022.4
						1598.6

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
MTD selection (%)						
U2 - shared alpha	66.3	1.8	0	0	0	31.9
U2 - dose-specific alpha	66.7	1.7	0	0	0	31.6
U2 - cycle-1 oracle	75.5	4.8	0.1	0	0	19.6
U3 - shared alpha	67.3	1.6	0	0	0	31.1
U3 - dose-specific alpha	67.5	1.7	0	0	0	30.8
U3 - cycle-1 oracle	75.5	5	0.2	0	0	19.3
OBD selection (%)						
U2 - shared alpha	7	0.4	0	0	0	92.6
U2 - dose-specific alpha	6.7	0.4	0	0	0	92.9
U2 - cycle-1 oracle	7.6	1.7	0.1	0	0	90.6
U3 - shared alpha	6.4	0.6	0	0	0	93
U3 - dose-specific alpha	6.5	0.4	0	0	0	93.1
U3 - cycle-1 oracle	7.8	1.5	0.1	0	0	90.6
Mean administrations						
U2 - shared alpha	11	3.5	0.3	0	0	-
U2 - dose-specific alpha	10.9	3.5	0.3	0	0	-
U2 - cycle-1 oracle	11	4.2	0.7	0.1	0	-
U3 - shared alpha	10.9	3.5	0.3	0	0	-
U3 - dose-specific alpha	10.9	3.5	0.3	0	0	-
U3 - cycle-1 oracle	11.1	4.2	0.7	0.1	0	-
Mean IPDE administrations						
U2 - shared alpha	2.9	2	0.1	0	0	-
U2 - dose-specific alpha	2.9	2	0.1	0	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.9	2	0.1	0	0	-
U3 - dose-specific alpha	2.9	2	0.1	0	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						531.3
U2 - dose-specific alpha						529.4
U2 - cycle-1 oracle						879.4
U3 - shared alpha						529.8
U3 - dose-specific alpha						528.8
U3 - cycle-1 oracle						879
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
MTD selection (%)						
U2 - shared alpha	0	0.1	0.3	45.9	53.7	0
U2 - dose-specific alpha	0	0	0.3	45.3	54.4	0
U2 - cycle-1 oracle	0	0.2	0.1	31	68.7	0
U3 - shared alpha	0	0	0.3	50.2	49.5	0
U3 - dose-specific alpha	0	0	0.2	49	50.8	0
U3 - cycle-1 oracle	0	0.2	0	34	65.8	0
OBD selection (%)						
U2 - shared alpha	0.6	0.9	2	60.6	35.9	0
U2 - dose-specific alpha	0.9	0.8	2.1	60.8	35.4	0
U2 - cycle-1 oracle	2.2	2	3.1	56.3	36.3	0.1
U3 - shared alpha	1.5	1.4	3.2	63.9	30	0
U3 - dose-specific alpha	1.3	1.5	2.5	63.6	31.1	0
U3 - cycle-1 oracle	3.7	3.6	4.5	58.1	30	0.1
Mean administrations						
U2 - shared alpha	3.1	3.2	3.8	13.5	6.4	-
U2 - dose-specific alpha	3.1	3.2	3.8	13.5	6.5	-
U2 - cycle-1 oracle	3.1	3.2	4	12.9	6.8	-
U3 - shared alpha	3.1	3.2	3.9	14.1	5.7	-
U3 - dose-specific alpha	3.1	3.2	3.9	14	5.9	-
U3 - cycle-1 oracle	3.1	3.2	4.1	13.3	6.4	-
Mean IPDE administrations						
U2 - shared alpha	0	2.4	1.2	4.7	1.9	-
U2 - dose-specific alpha	0	2.5	1.2	4.7	2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0	2.5	1.2	4.9	1.7	-
U3 - dose-specific alpha	0	2.5	1.2	4.8	1.8	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1088.5
U2 - dose-specific alpha						1088.2
U2 - cycle-1 oracle						1663.2
U3 - shared alpha						1086.9
U3 - dose-specific alpha						1087
U3 - cycle-1 oracle						1663.2

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
MTD selection (%)						
U2 - shared alpha	3.3	44	49	3.7	0	0
U2 - dose-specific alpha	3.4	43.1	49.9	3.6	0	0
U2 - cycle-1 oracle	2.5	35.6	51.8	9.8	0.3	0
U3 - shared alpha	3.6	43.8	49	3.6	0	0
U3 - dose-specific alpha	3.8	43.9	48.7	3.6	0	0
U3 - cycle-1 oracle	2	36.7	50.8	10.2	0.3	0
OBD selection (%)						
U2 - shared alpha	27.5	37.7	31.9	2.1	0	0.8
U2 - dose-specific alpha	27.6	38	31.6	1.8	0	1
U2 - cycle-1 oracle	27.6	32.5	33.5	5	0	1.4
U3 - shared alpha	32.4	38	27	1.7	0	0.9
U3 - dose-specific alpha	33.3	37.7	26.4	1.4	0	1.2
U3 - cycle-1 oracle	33.1	31.8	28.4	5.3	0	1.4
Mean administrations						
U2 - shared alpha	9.3	10.6	8.1	1.7	0.1	-
U2 - dose-specific alpha	9.2	10.6	8.1	1.8	0.1	-
U2 - cycle-1 oracle	8.9	9.3	8.7	2.6	0.4	-
U3 - shared alpha	9.8	10.8	7.6	1.6	0.1	-
U3 - dose-specific alpha	9.8	10.8	7.7	1.5	0.1	-
U3 - cycle-1 oracle	9.4	9.4	8.3	2.5	0.3	-
Mean IPDE administrations						
U2 - shared alpha	2.4	4.5	2.4	0.7	0	-
U2 - dose-specific alpha	2.4	4.5	2.4	0.6	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.6	4.6	2.2	0.6	0	-
U3 - dose-specific alpha	2.6	4.6	2.2	0.6	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1100
U2 - dose-specific alpha						1099.3
U2 - cycle-1 oracle						1656.4
U3 - shared alpha						1094.2
U3 - dose-specific alpha						1094.7
U3 - cycle-1 oracle						1656.4

Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
MTD selection (%)						
U2 - shared alpha	18.4	42	34.5	4.8	0	0.3
U2 - dose-specific alpha	18.8	40.3	34.7	5.9	0	0.3
U2 - cycle-1 oracle	14.1	32.4	41.8	11.2	0.2	0.3
U3 - shared alpha	18.7	41.3	34.7	5	0	0.3
U3 - dose-specific alpha	18.4	41.9	34.2	5.2	0	0.3
U3 - cycle-1 oracle	13.7	33.3	41	11.5	0.2	0.3
OBD selection (%)						
U2 - shared alpha	22.9	35.9	25.7	2.5	0	13
U2 - dose-specific alpha	23.2	34.8	25.4	3.2	0	13.4
U2 - cycle-1 oracle	19.3	29.7	32.2	5.7	0.1	13
U3 - shared alpha	23.9	37	23.4	2.2	0	13.5
U3 - dose-specific alpha	25.1	36.9	22.6	2.5	0	12.9
U3 - cycle-1 oracle	21.1	33.1	27.3	5.6	0.1	12.8
Mean administrations						
U2 - shared alpha	9.5	9.6	7	2.1	0.2	-
U2 - dose-specific alpha	9.5	9.7	7	2.1	0.1	-
U2 - cycle-1 oracle	8.2	8.7	8.1	3	0.4	-
U3 - shared alpha	9.8	9.7	6.8	1.9	0.1	-
U3 - dose-specific alpha	9.8	9.8	6.7	1.9	0.1	-
U3 - cycle-1 oracle	8.5	8.8	7.9	2.9	0.4	-
Mean IPDE administrations						
U2 - shared alpha	2.4	4.5	2.1	0.8	0	-
U2 - dose-specific alpha	2.4	4.6	2.1	0.8	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.6	4.6	2	0.8	0	-
U3 - dose-specific alpha	2.6	4.6	2	0.8	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1011.4
U2 - dose-specific alpha						1012.6
U2 - cycle-1 oracle						1577.2
U3 - shared alpha						1008.1
U3 - dose-specific alpha						1009.7
U3 - cycle-1 oracle						1576.4

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
MTD selection (%)						
U2 - shared alpha	6.6	29.8	52.2	11	0.4	0
U2 - dose-specific alpha	6.9	29.1	51.9	11.8	0.3	0
U2 - cycle-1 oracle	5.2	19.8	52.5	20.1	2.4	0
U3 - shared alpha	6.2	29.5	54	10.1	0.2	0
U3 - dose-specific alpha	6.3	30.1	53.4	10	0.2	0
U3 - cycle-1 oracle	5.3	20.5	53.1	19.3	1.8	0
OBD selection (%)						
U2 - shared alpha	14	30.7	43.1	6.4	0.1	5.7
U2 - dose-specific alpha	14.5	29.3	43	7	0.1	6.1
U2 - cycle-1 oracle	12.8	25	45.8	10.3	0.4	5.7
U3 - shared alpha	17.2	30.9	41.5	4.9	0	5.5
U3 - dose-specific alpha	17.8	31.2	40.4	4.9	0.1	5.6
U3 - cycle-1 oracle	16.3	26.5	42.5	8.7	0.4	5.6
Mean administrations						
U2 - shared alpha	7.1	9.2	9.8	2.9	0.4	-
U2 - dose-specific alpha	7	9.2	9.8	2.9	0.4	-
U2 - cycle-1 oracle	6.3	8.2	10.1	3.9	0.8	-
U3 - shared alpha	7.4	9.3	9.7	2.6	0.3	-
U3 - dose-specific alpha	7.3	9.4	9.6	2.7	0.3	-
U3 - cycle-1 oracle	6.6	8.3	10	3.7	0.7	-
Mean IPDE administrations						
U2 - shared alpha	1.6	4.3	2.9	1.1	0.1	-
U2 - dose-specific alpha	1.5	4.3	2.9	1.1	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.7	4.4	2.8	1	0.1	-
U3 - dose-specific alpha	1.7	4.4	2.8	1	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1063.8
U2 - dose-specific alpha						1062.5
U2 - cycle-1 oracle						1630.2
U3 - shared alpha						1060.2
U3 - dose-specific alpha						1058.7
U3 - cycle-1 oracle						1629.4

Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
MTD selection (%)						
U2 - shared alpha	59.1	32.5	5.9	0.1	0	2.4
U2 - dose-specific alpha	59.8	31.3	6.2	0.2	0	2.5
U2 - cycle-1 oracle	44.9	37.3	14.3	1.6	0	1.9
U3 - shared alpha	59.1	31.9	6.7	0.1	0	2.2
U3 - dose-specific alpha	58.9	31.9	6.8	0.1	0	2.3
U3 - cycle-1 oracle	45	36	15.6	1.5	0	1.9
OBD selection (%)						
U2 - shared alpha	15.6	32.2	2.5	0.1	0	49.6
U2 - dose-specific alpha	15.1	31.1	3.4	0.1	0	50.3
U2 - cycle-1 oracle	10.8	41.6	6.8	0.2	0	40.6
U3 - shared alpha	15.9	31.8	2.6	0	0	49.7
U3 - dose-specific alpha	15.2	32.3	2.6	0	0	49.9
U3 - cycle-1 oracle	10.9	41	7.1	0.1	0	40.9
Mean administrations						
U2 - shared alpha	10.6	10	2.6	0.6	0	-
U2 - dose-specific alpha	10.6	10	2.6	0.6	0	-
U2 - cycle-1 oracle	8.8	10.9	3.9	1.1	0.1	-
U3 - shared alpha	10.7	9.9	2.6	0.5	0	-
U3 - dose-specific alpha	10.7	10	2.5	0.6	0	-
U3 - cycle-1 oracle	8.9	11	3.7	1	0.1	-
Mean IPDE administrations						
U2 - shared alpha	2.7	4.5	0.8	0.3	0	-
U2 - dose-specific alpha	2.7	4.5	0.8	0.3	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.7	4.4	0.8	0.2	0	-
U3 - dose-specific alpha	2.7	4.5	0.8	0.2	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						854
U2 - dose-specific alpha						853.4
U2 - cycle-1 oracle						1380
U3 - shared alpha						848.9
U3 - dose-specific alpha						852
U3 - cycle-1 oracle						1377.5

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
MTD selection (%)						
U2 - shared alpha	42.1	50	5.4	0	0	2.5
U2 - dose-specific alpha	41.5	51.4	4.8	0	0	2.3
U2 - cycle-1 oracle	26.9	60.1	10.1	0.8	0	2.1
U3 - shared alpha	42.5	51.4	3.6	0	0	2.5
U3 - dose-specific alpha	41.1	53.2	3	0	0	2.7
U3 - cycle-1 oracle	27.5	60.3	9.5	0.7	0	2
OBD selection (%)						
U2 - shared alpha	69.2	26.2	1.8	0	0	2.8
U2 - dose-specific alpha	67.7	27.9	1.5	0	0	2.9
U2 - cycle-1 oracle	64.3	29.9	3.3	0.2	0	2.3
U3 - shared alpha	72.3	24	0.9	0	0	2.8
U3 - dose-specific alpha	72	24.4	0.8	0	0	2.8
U3 - cycle-1 oracle	66.3	28.7	2.5	0.1	0	2.4
Mean administrations						
U2 - shared alpha	19.1	8.6	1.5	0.2	0	-
U2 - dose-specific alpha	19	8.7	1.5	0.2	0	-
U2 - cycle-1 oracle	17.8	8.7	2.5	0.5	0	-
U3 - shared alpha	20	8.2	1.2	0.1	0	-
U3 - dose-specific alpha	20.1	8.2	1.1	0.1	0	-
U3 - cycle-1 oracle	18.5	8.5	2.1	0.4	0	-
Mean IPDE administrations						
U2 - shared alpha	4.7	3.2	0.5	0.1	0	-
U2 - dose-specific alpha	4.7	3.2	0.5	0.1	0	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	5.1	3.1	0.4	0	0	-
U3 - dose-specific alpha	5.1	3.1	0.3	0	0	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1160.6
U2 - dose-specific alpha						1158.7
U2 - cycle-1 oracle						1641.4
U3 - shared alpha						1157.8
U3 - dose-specific alpha						1156.9
U3 - cycle-1 oracle						1640.7
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
MTD selection (%)						
U2 - shared alpha	0.4	2.8	22.3	58.7	15.8	0
U2 - dose-specific alpha	0.4	2.8	21.7	60.4	14.7	0
U2 - cycle-1 oracle	0.5	1.3	14.5	56.4	27.3	0
U3 - shared alpha	0.4	3.3	21.3	60.2	14.8	0
U3 - dose-specific alpha	0.4	3.1	21.2	61.4	13.9	0
U3 - cycle-1 oracle	0.5	1.5	14.4	57.9	25.7	0
OBD selection (%)						
U2 - shared alpha	15.9	13.6	16.1	45.3	8.6	0.5
U2 - dose-specific alpha	15.9	12.3	17	46.3	8	0.5
U2 - cycle-1 oracle	16	12.9	12.8	45.5	12.3	0.5
U3 - shared alpha	22.2	15.3	15.2	39.9	6.9	0.5
U3 - dose-specific alpha	21	15.7	15.5	40.2	7.1	0.5
U3 - cycle-1 oracle	24.3	15.7	11.3	38.6	9.6	0.5
Mean administrations						
U2 - shared alpha	4.5	5.6	6.6	10.1	3.2	-
U2 - dose-specific alpha	4.6	5.5	6.5	10.2	3.1	-
U2 - cycle-1 oracle	4.3	5	6.1	10.3	4.2	-
U3 - shared alpha	4.9	5.7	6.5	10.1	2.8	-
U3 - dose-specific alpha	4.9	5.8	6.5	10	2.7	-
U3 - cycle-1 oracle	4.5	5.1	6.1	10.3	3.9	-
Mean IPDE administrations						
U2 - shared alpha	0.6	3.2	2.1	3.3	1	-
U2 - dose-specific alpha	0.7	3.2	2.1	3.4	1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.8	3.3	2.1	3.4	0.9	-
U3 - dose-specific alpha	0.8	3.3	2.1	3.4	0.9	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1075.7
U2 - dose-specific alpha						1076
U2 - cycle-1 oracle						1659.7
U3 - shared alpha						1072.3
U3 - dose-specific alpha						1069.6
U3 - cycle-1 oracle						1659.7

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
MTD selection (%)						
U2 - shared alpha	0.1	40.5	30.8	24.1	4.5	0
U2 - dose-specific alpha	0.1	40.1	31.7	23.5	4.6	0
U2 - cycle-1 oracle	0	25.6	25.4	31.2	17.8	0
U3 - shared alpha	0.1	47	28.7	21.3	2.9	0
U3 - dose-specific alpha	0.1	46.5	29.6	20.4	3.4	0
U3 - cycle-1 oracle	0	29.9	25.2	30.3	14.6	0
OBD selection (%)						
U2 - shared alpha	2.3	45.1	31.3	19.3	2	0
U2 - dose-specific alpha	1.7	45	32.8	18.5	2	0
U2 - cycle-1 oracle	4.6	35.7	28.4	25.2	6.1	0
U3 - shared alpha	3.8	50.8	27.9	16.6	0.9	0
U3 - dose-specific alpha	3.8	50.2	28.7	16.4	0.9	0
U3 - cycle-1 oracle	6.4	42.5	26.7	20.1	4.3	0
Mean administrations						
U2 - shared alpha	3.9	14.3	7.6	3.7	0.5	-
U2 - dose-specific alpha	3.9	14.3	7.6	3.6	0.5	-
U2 - cycle-1 oracle	3.8	11.6	7.4	5.5	1.7	-
U3 - shared alpha	4.1	15.3	7	3.2	0.4	-
U3 - dose-specific alpha	4.1	15.2	7.1	3.2	0.4	-
U3 - cycle-1 oracle	3.8	12.4	7.2	5.2	1.4	-
Mean IPDE administrations						
U2 - shared alpha	0.3	5.4	2.3	1.1	0.2	-
U2 - dose-specific alpha	0.3	5.4	2.4	1.1	0.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.4	5.8	2.2	1	0.1	-
U3 - dose-specific alpha	0.4	5.8	2.2	1	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1139
U2 - dose-specific alpha						1137.8
U2 - cycle-1 oracle						1664.1
U3 - shared alpha						1135.4
U3 - dose-specific alpha						1135.8
U3 - cycle-1 oracle						1664.1
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
MTD selection (%)						
U2 - shared alpha	0.2	0.2	10.5	50.6	38.5	0
U2 - dose-specific alpha	0.1	0.4	10.2	52.6	36.7	0
U2 - cycle-1 oracle	0.1	0.4	7	35.4	57.1	0
U3 - shared alpha	0.1	0.6	10.5	54.8	34	0
U3 - dose-specific alpha	0.1	0.6	10.1	54.9	34.3	0
U3 - cycle-1 oracle	0.1	0.3	6.9	41.1	51.6	0
OBD selection (%)						
U2 - shared alpha	11.6	13.7	33.9	31.5	9.1	0.2
U2 - dose-specific alpha	10.8	14	35.2	31.5	8.3	0.2
U2 - cycle-1 oracle	13.4	14.2	33.2	27.2	11.8	0.2
U3 - shared alpha	15.5	17.9	34	27.1	5.4	0.1
U3 - dose-specific alpha	15.3	16.2	35	27.4	6	0.1
U3 - cycle-1 oracle	18.9	16.9	32.7	23.1	8.2	0.2
Mean administrations						
U2 - shared alpha	3.5	4.5	8.9	8.9	4.2	-
U2 - dose-specific alpha	3.5	4.6	8.9	8.9	4.1	-
U2 - cycle-1 oracle	3.5	4.4	8.6	8.5	5	-
U3 - shared alpha	3.6	4.7	9.4	8.7	3.6	-
U3 - dose-specific alpha	3.7	4.8	9.3	8.6	3.5	-
U3 - cycle-1 oracle	3.5	4.5	8.9	8.5	4.5	-
Mean IPDE administrations						
U2 - shared alpha	0.2	2.9	2.9	3.4	1.4	-
U2 - dose-specific alpha	0.2	3	2.9	3.3	1.4	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.3	3	3.1	3.3	1.2	-
U3 - dose-specific alpha	0.3	3.1	3.1	3.3	1.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1054.2
U2 - dose-specific alpha						1052.7
U2 - cycle-1 oracle						1662.7
U3 - shared alpha						1050.1
U3 - dose-specific alpha						1048.7
U3 - cycle-1 oracle						1662.7

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
MTD selection (%)						
U2 - shared alpha	0	0	11	59.9	29.1	0
U2 - dose-specific alpha	0	0	10.2	61.6	28.2	0
U2 - cycle-1 oracle	0	0.3	7.1	45.9	46.7	0
U3 - shared alpha	0	0.1	10.2	64	25.7	0
U3 - dose-specific alpha	0	0	9.6	64.3	26.1	0
U3 - cycle-1 oracle	0	0.4	6.2	49.6	43.8	0
OBD selection (%)						
U2 - shared alpha	7.8	6.1	13.6	51.8	20.6	0.1
U2 - dose-specific alpha	8.8	5.7	13.2	51.6	20.6	0.1
U2 - cycle-1 oracle	8.1	5.8	11.3	44.9	29.7	0.2
U3 - shared alpha	12	6.6	13.3	50.3	17.7	0.1
U3 - dose-specific alpha	12.6	7.4	12.1	50.3	17.5	0.1
U3 - cycle-1 oracle	13	8.1	10.5	44.8	23.4	0.2
Mean administrations						
U2 - shared alpha	3.3	3.8	6.3	11.2	5.4	-
U2 - dose-specific alpha	3.3	3.8	6.3	11.2	5.3	-
U2 - cycle-1 oracle	3.4	3.7	6.1	10.5	6.3	-
U3 - shared alpha	3.5	3.9	6.4	11.5	4.8	-
U3 - dose-specific alpha	3.5	3.9	6.4	11.4	4.8	-
U3 - cycle-1 oracle	3.4	3.8	6.2	10.6	5.9	-
Mean IPDE administrations						
U2 - shared alpha	0.1	2.7	1.9	4	1.5	-
U2 - dose-specific alpha	0.2	2.7	1.9	4	1.5	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.2	2.8	1.9	4	1.3	-
U3 - dose-specific alpha	0.2	2.8	1.9	4.1	1.3	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1087.4
U2 - dose-specific alpha						1087.7
U2 - cycle-1 oracle						1662.8
U3 - shared alpha						1084.1
U3 - dose-specific alpha						1083
U3 - cycle-1 oracle						1662.8

Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
MTD selection (%)						
U2 - shared alpha	2.2	12.7	52.7	27	5.4	0
U2 - dose-specific alpha	2	13	52.7	26.9	5.4	0
U2 - cycle-1 oracle	0.9	10.3	37.5	37.3	14	0
U3 - shared alpha	2	13.6	51.7	27.6	5.1	0
U3 - dose-specific alpha	1.8	13.5	51.4	28.5	4.8	0
U3 - cycle-1 oracle	0.9	9.8	38.1	39.3	11.9	0
OBD selection (%)						
U2 - shared alpha	12.6	15.9	49.3	14.8	2.4	5
U2 - dose-specific alpha	12.8	16.6	48.8	14.9	2.2	4.7
U2 - cycle-1 oracle	12.3	13.5	44.3	21.8	4.5	3.6
U3 - shared alpha	16.3	18.9	44.3	13.5	2	5
U3 - dose-specific alpha	16.5	17.9	44.7	14.2	1.8	4.9
U3 - cycle-1 oracle	17.1	15.4	41.6	19.5	2.5	3.9
Mean administrations						
U2 - shared alpha	5.4	7	10.3	5.4	1.5	-
U2 - dose-specific alpha	5.4	7	10.2	5.4	1.5	-
U2 - cycle-1 oracle	5	6.3	9.7	6.2	2.5	-
U3 - shared alpha	5.7	7.2	10.2	5.3	1.3	-
U3 - dose-specific alpha	5.7	7.2	10.2	5.2	1.3	-
U3 - cycle-1 oracle	5.3	6.4	9.7	6	2.2	-
Mean IPDE administrations						
U2 - shared alpha	1	3.8	3.2	2	0.5	-
U2 - dose-specific alpha	1	3.8	3.2	2	0.5	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	1.2	3.9	3.2	1.9	0.5	-
U3 - dose-specific alpha	1.2	3.9	3.2	1.9	0.4	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1044.9
U2 - dose-specific alpha						1044.9
U2 - cycle-1 oracle						1647.1
U3 - shared alpha						1041.7
U3 - dose-specific alpha						1042.2
U3 - cycle-1 oracle						1646.2

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy and toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; blue rows = dose-specific alpha, green rows = cycle-1 oracle.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
MTD selection (%)						
U2 - shared alpha	0.1	31.3	51.4	15.1	2.1	0
U2 - dose-specific alpha	0.1	31.5	50.4	16.1	1.9	0
U2 - cycle-1 oracle	0.1	20.9	42.3	28.1	8.6	0
U3 - shared alpha	0.1	37.6	47.5	13.5	1.3	0
U3 - dose-specific alpha	0.1	37.6	48	12.8	1.5	0
U3 - cycle-1 oracle	0.1	23.5	43.6	25.7	7.1	0
OBD selection (%)						
U2 - shared alpha	6.4	47.8	36.5	8.2	1	0.1
U2 - dose-specific alpha	6.7	48.2	34.9	9	1.1	0.1
U2 - cycle-1 oracle	9	41.6	30.7	15	3.6	0.1
U3 - shared alpha	9.1	53.1	30.2	6.8	0.7	0.1
U3 - dose-specific alpha	8.3	53.7	30.7	6.5	0.7	0.1
U3 - cycle-1 oracle	12.7	45.9	28.5	10.5	2.3	0.1
Mean administrations						
U2 - shared alpha	4.3	13.7	8.7	2.8	0.5	-
U2 - dose-specific alpha	4.4	13.8	8.4	2.8	0.5	-
U2 - cycle-1 oracle	4.5	11.5	8.4	4.2	1.4	-
U3 - shared alpha	4.7	14.7	7.9	2.3	0.4	-
U3 - dose-specific alpha	4.7	14.7	7.9	2.3	0.4	-
U3 - cycle-1 oracle	4.5	12.4	8.1	3.8	1.2	-
Mean IPDE administrations						
U2 - shared alpha	0.5	5.3	2.5	0.9	0.1	-
U2 - dose-specific alpha	0.5	5.3	2.4	0.9	0.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	0.6	5.7	2.2	0.8	0.1	-
U3 - dose-specific alpha	0.6	5.7	2.2	0.7	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1143.8
U2 - dose-specific alpha						1144.3
U2 - cycle-1 oracle						1663.2
U3 - shared alpha						1136.5
U3 - dose-specific alpha						1137.8
U3 - cycle-1 oracle						1663.2
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
MTD selection (%)						
U2 - shared alpha	31.8	38.1	22.1	7	0.8	0.2
U2 - dose-specific alpha	31.5	39.4	20.9	7.1	0.9	0.2
U2 - cycle-1 oracle	22.1	33.3	26.3	13	5.1	0.2
U3 - shared alpha	31	39.8	21.9	6.4	0.7	0.2
U3 - dose-specific alpha	31.8	39.3	21.7	6.2	0.8	0.2
U3 - cycle-1 oracle	21.9	33.3	26.6	13.9	4.1	0.2
OBD selection (%)						
U2 - shared alpha	15.1	39.7	12.8	3.5	0.4	28.5
U2 - dose-specific alpha	15.8	40.4	11.5	3.6	0.5	28.2
U2 - cycle-1 oracle	13	40.8	14.3	6.7	1.8	23.4
U3 - shared alpha	17.6	40.2	10.3	2.7	0.3	28.9
U3 - dose-specific alpha	17.4	40.3	10.2	2.7	0.3	29.1
U3 - cycle-1 oracle	15.9	42.6	10.4	6.7	1.2	23.2
Mean administrations						
U2 - shared alpha	8.8	10.2	4.8	2.4	0.5	-
U2 - dose-specific alpha	8.8	10.3	4.8	2.4	0.5	-
U2 - cycle-1 oracle	7.7	10	5.4	3.3	1.1	-
U3 - shared alpha	9.1	10.3	4.6	2.3	0.4	-
U3 - dose-specific alpha	9.1	10.3	4.5	2.3	0.5	-
U3 - cycle-1 oracle	8	9.9	5.3	3.2	1	-
Mean IPDE administrations						
U2 - shared alpha	2.1	5	1.5	1	0.2	-
U2 - dose-specific alpha	2.1	5	1.5	0.9	0.2	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.2	5	1.5	0.9	0.2	-
U3 - dose-specific alpha	2.2	5	1.4	0.9	0.2	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						935.3
U2 - dose-specific alpha						938
U2 - cycle-1 oracle						1521.4
U3 - shared alpha						930.9
U3 - dose-specific alpha						930.8
U3 - cycle-1 oracle						1523

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy with cycle-1 oracle - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
MTD selection (%)						
U2 - shared alpha	25.9	48	22.7	3.1	0.3	0
U2 - dose-specific alpha	25.4	48.2	23.3	2.8	0.3	0
U2 - cycle-1 oracle	19.6	38.9	30.6	8.5	2.4	0
U3 - shared alpha	25.3	47.6	23.1	3.7	0.3	0
U3 - dose-specific alpha	24.5	48.3	23.6	3.3	0.3	0
U3 - cycle-1 oracle	19.3	39.3	29.8	9.7	1.9	0
OBD selection (%)						
U2 - shared alpha	15.1	48.9	11.2	1	0	23.8
U2 - dose-specific alpha	15.2	48.9	11.1	1	0	23.8
U2 - cycle-1 oracle	13.7	47.9	13.1	3.2	0.6	21.5
U3 - shared alpha	17.5	47.9	10.1	0.9	0	23.6
U3 - dose-specific alpha	17.1	48.2	10.8	1	0	22.9
U3 - cycle-1 oracle	16	48.3	11.4	2.5	0.6	21.2
Mean administrations						
U2 - shared alpha	8.6	11.7	4.9	2.1	0.4	-
U2 - dose-specific alpha	8.6	11.7	4.9	2.1	0.4	-
U2 - cycle-1 oracle	7.6	11.2	5.6	2.9	0.8	-
U3 - shared alpha	8.9	11.8	4.7	2	0.4	-
U3 - dose-specific alpha	8.9	11.7	4.7	2	0.4	-
U3 - cycle-1 oracle	8	11.2	5.4	2.8	0.8	-
Mean IPDE administrations						
U2 - shared alpha	2	5.6	1.5	0.9	0.1	-
U2 - dose-specific alpha	2	5.6	1.5	0.9	0.1	-
U2 - cycle-1 oracle	0	0	0	0	0	-
U3 - shared alpha	2.2	5.7	1.5	0.8	0.1	-
U3 - dose-specific alpha	2.2	5.7	1.4	0.8	0.1	-
U3 - cycle-1 oracle	0	0	0	0	0	-
Mean trial duration (days)						
U2 - shared alpha						Days
U2 - dose-specific alpha						960
U2 - cycle-1 oracle						959.2
U3 - shared alpha						1557.6
U3 - dose-specific alpha						959.7
U3 - cycle-1 oracle						958.3
						1557.3